# aiDAPTIV+ GUI

# Pro Suite User guide

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# 1. ENVIRONMENT PREPARATION

To ensure optimal performance and compatibility, the following system requirements must be met:

- **Operating System:** Any Linux-based operating system
- **Container Platform:** Docker
- **GPU Support:** NVIDIA GPU with NVIDIA Container Toolkit installed
- **Storage:** aiDAPTIV drive properly installed and connected

# 2. OVERVIEW

MiPhi aiDAPTIV GUI is an advanced, user-friendly platform designed to streamline the AI development lifecycle. It provides an intuitive graphical interface for managing datasets, models, fine-tuning processes, and inference workflows.

## 2.1 Feature Overview

The MiPhi aiDAPTIV GUI includes the following core functionalities:

- Login
- System Dashboard
- Datasets
- Models
- Fine-tuning
- Monitor
- Inference
- Logs
- History

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# 3. LOGIN

The **MiPhi Login** is the secure gateway to your MiPhi environment. All user accounts—including role-based credentials—are provisioned and managed by a default admin user. Once the admin has created your username and initial password, simply enter those credentials here to access the system. This centralized control ensures that only authorized personnel can log in, while still allowing the admin to delegate additional user-creation or management tasks as needed.

## 3.1 Sign In to MiPhi

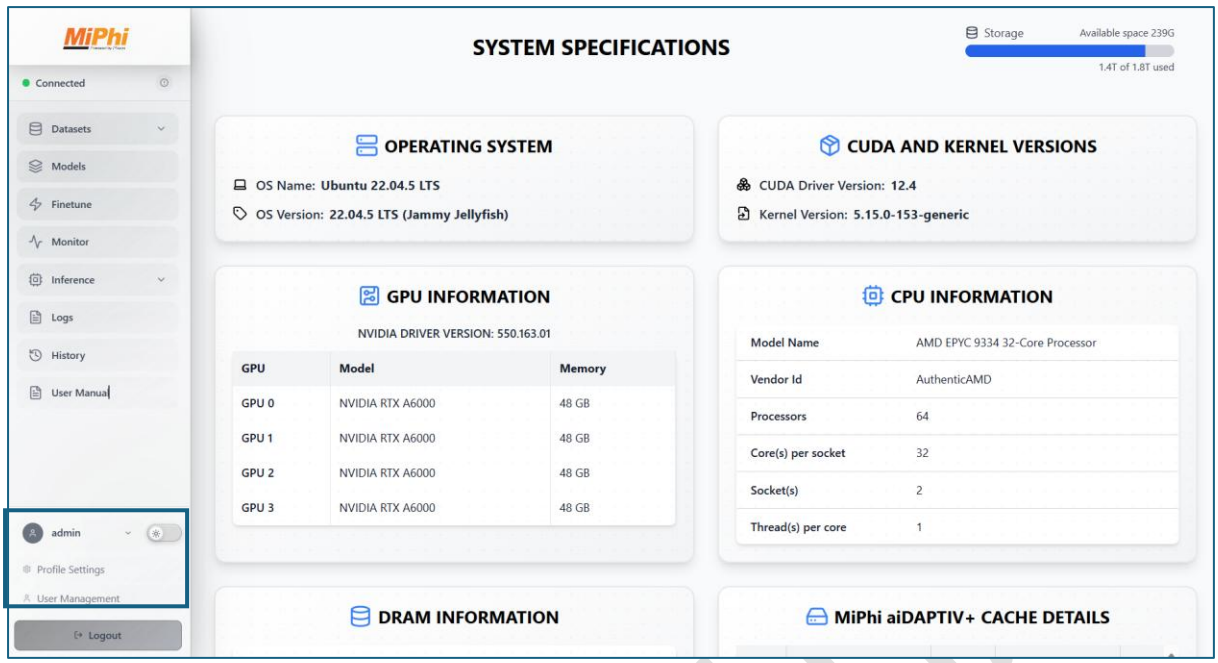
- Enter the **default username and password** provided to authenticate.
    - Default Email: admin@miphi.in
    - Default Password: Admin1234
- Use the "👁" icon in the password field to toggle visibility and avoid typos.




## 3.2 Profile Settings & User Management

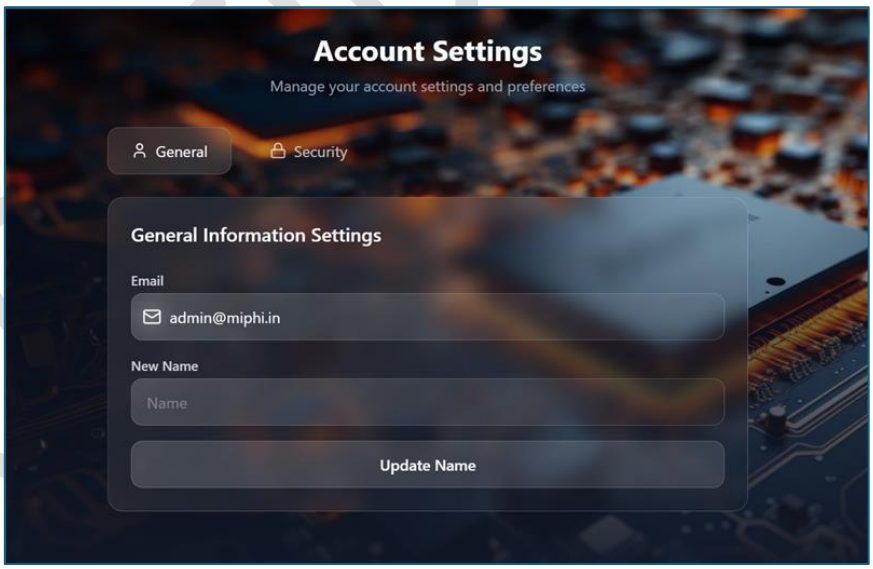
- Click the user-icon in the left-bottom to open the dropdown.
- Select Profile Settings to manage your profile.
- The Profile Settings page has two tabs—General and Security—for profile and password management.

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### 3.2.1 General Settings

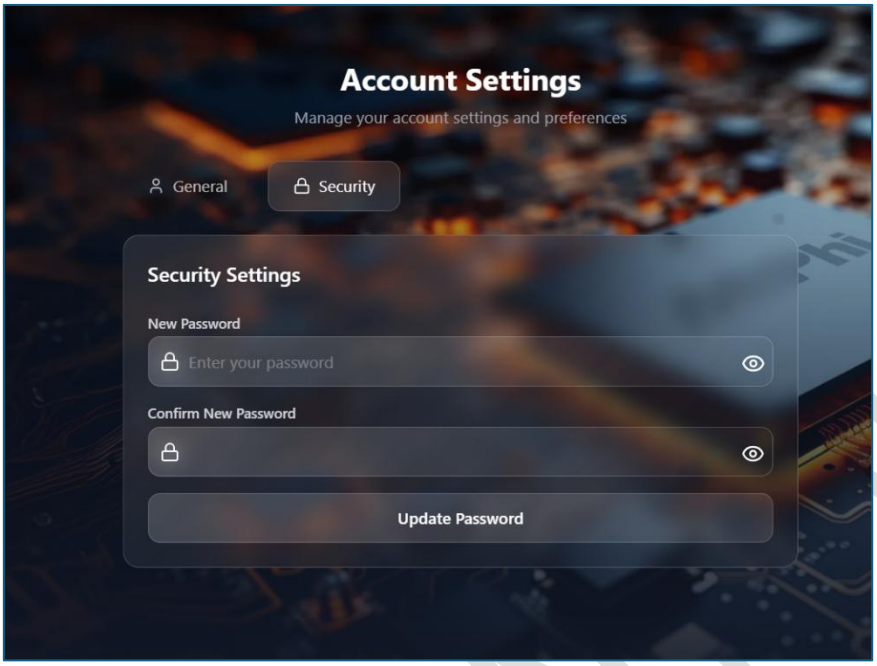
- Email (read-only): Displays your registered address.
- New Name field: Enter a new display name.
- Update Name button: Apply and save your name change instantly.




### 3.2.2 Security Settings

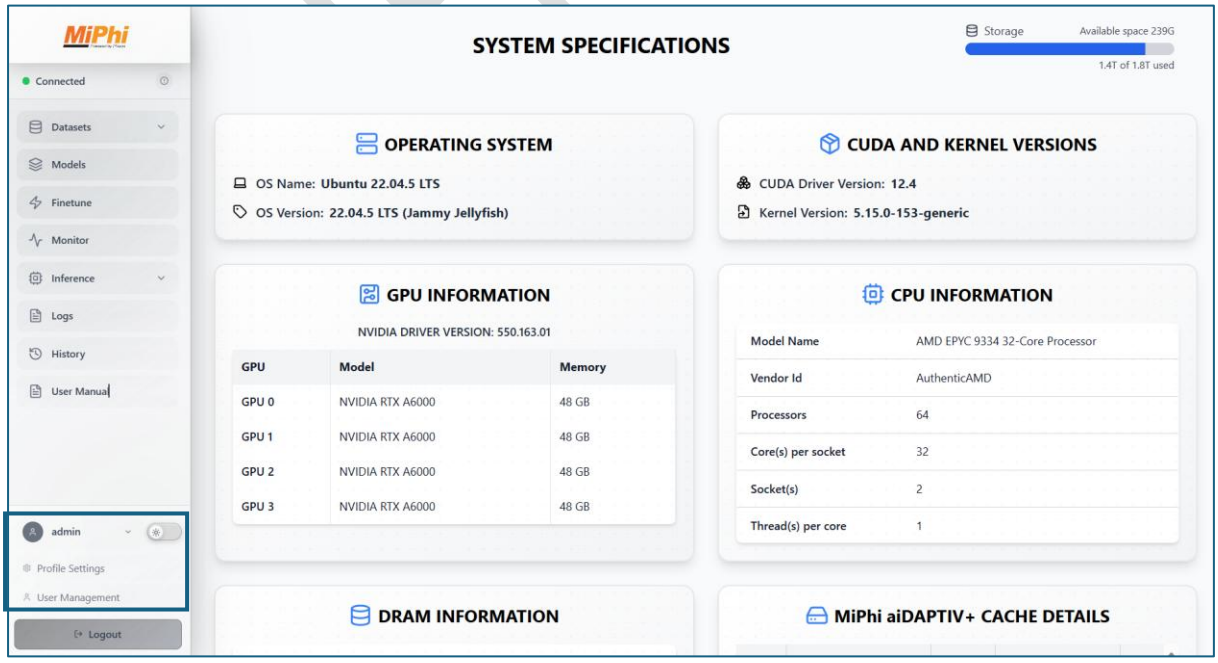
- Switch to the Security tab under Account Settings.
- Enter your New Password, then repeat it in Confirm New Password.
- Use the “ 👁 ” icon to toggle visibility and avoid typos.

- Click Update Password to save your changes.



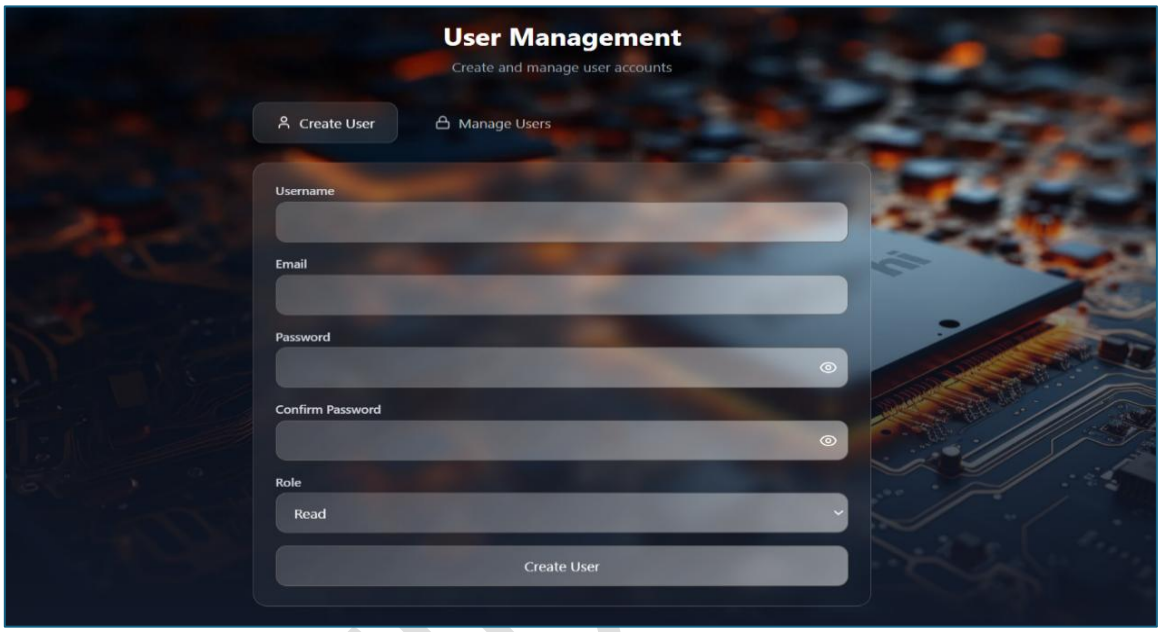

## 3.3 User Management (Admin Only)

- Accessible via the User Management section as Create User and Manage Users.
- Empowers the admin to provision new accounts or review and delete existing ones.
- Ensures fine-grained control over who can log in and what role they receive.



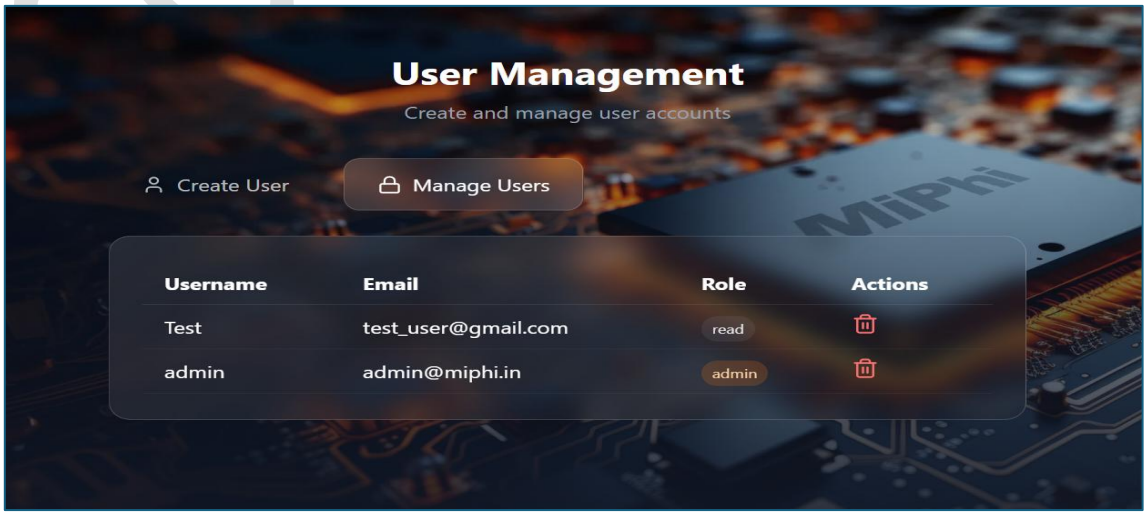
### 3.3.1 Create User

- Username, Email, Password and Confirm Password fields collect the new account details.
- The Role dropdown assigns permissions (e.g. “read” , “write” , “admin”).
- Click Create User to provision the account and send the credentials to the user.



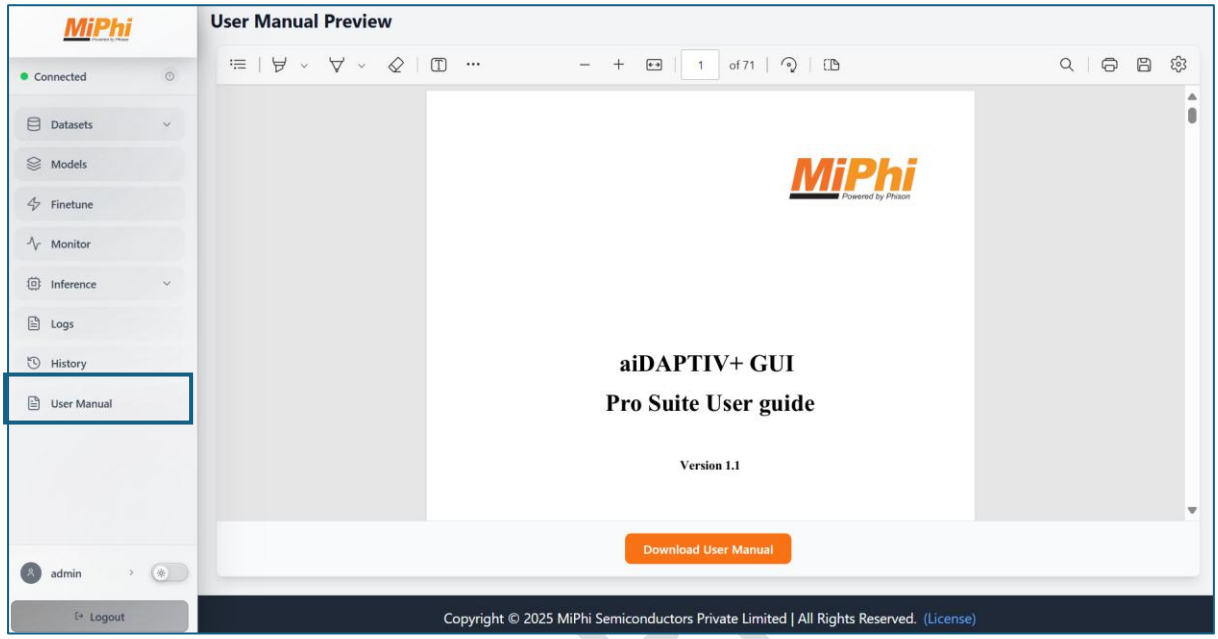

### 3.3.2 Manage Users

- Displays a table of all current users with Username, Email, and Role.
- The 🗑 Delete icon removes an account permanently (with confirmation).
- Use this view to audit active users and revoke access instantly.



## 3.4 User Manual

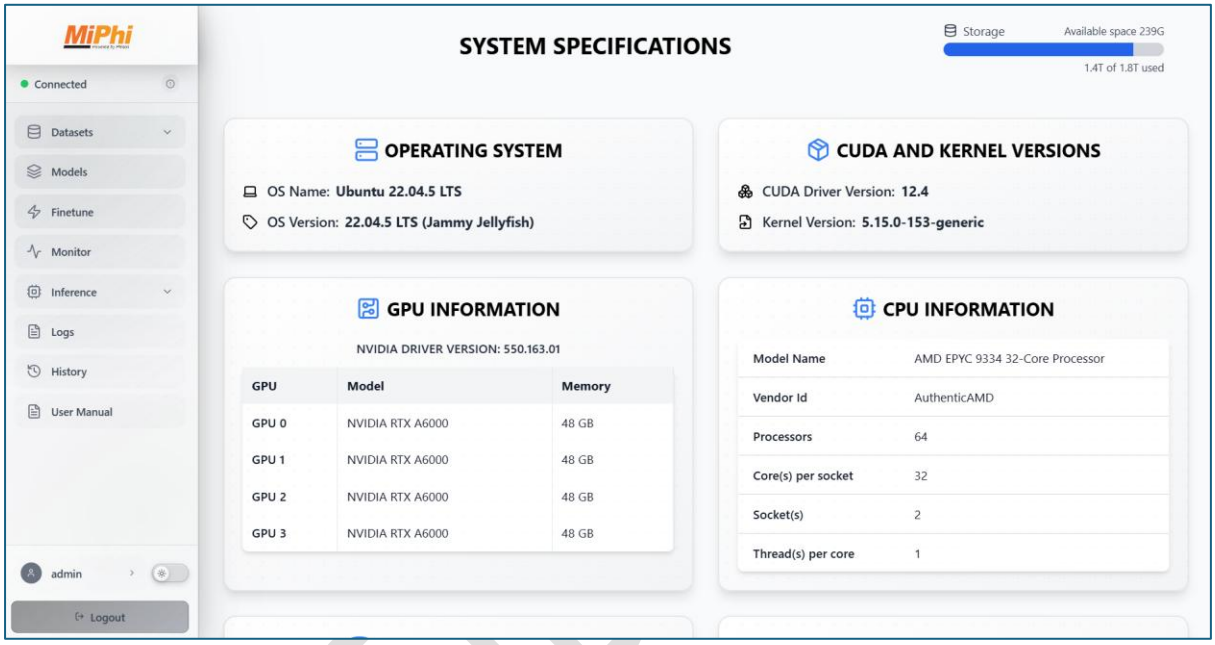
- You can preview the user manual by clicking the **User Manual** located on the side navigation bar, as shown below:




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# 4. SYSTEM DASHBOARD

The **MiPhi System Dashboard** delivers a unified and detailed view of the system's hardware and software specifications. Designed for users to facilitates real-time monitoring and management of compute nodes within high-performance environments. This dashboard dynamically reflects the current server configuration, ensuring accurate and up-to-date information for performance analysis, troubleshooting, and capacity planning.




## 4.1 Operating System Overview

- **Linux Distribution:** Shows the full name of the Linux OS currently running. ***Example***: *Ubuntu 22.04.5 LTS*
- **Release Version & Codename:** Indicates the release version and associated codename for easy identification.
  ***Example:*** *22.04.5 LTS (Jammy Jellyfish)*

## 4.2 CUDA & Kernel Information

- **CUDA Driver Version:** Displays the installed CUDA driver version used for GPU-accelerated applications.
  ***Example:*** *CUDA Driver Version: 12.4*
- **Linux Kernel Version:** Shows the current kernel version running on the system. ***Example***: *Kernel Version: 5.15.0-138-generic*

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## 4.3 GPU Information

- **Installed GPUs:** Lists each GPU model available in the system.
  ***Example:*** *NVIDIA RTX A6000*
- **Memory per GPU:** Shows how much memory each GPU has.
  ***Example:*** *48 GB per unit*
- **NVIDIA Driver Version:** Indicates the active driver version used by all GPUs.
  ***Example:*** *Driver Version: 550.120*

## 4.4 CPU Details

- **Processor Model & Architecture:** Displays the CPU vendor, model, and core architecture.
  ***Example****: AMD EPYC 9334 32-Core Processor (AuthenticAMD)*
- **Core & Thread Configuration:** Shows how many cores and threads are present and how they're distributed across sockets.
  ***Example:*** *128 threads, 32 cores/socket × 2 sockets*
- **Per-Core Threading:** Provides thread-level visibility useful for tuning performance or virtualization.

## 4.5 DRAM (Memory) Specifications

- **Memory Type & Total Capacity:** Indicates the type (e.g., DDR5) and total installed memory.
  ***Example****: DDR5, Total DRAM Size: 2112 GB*
- **Number of DIMMs:** Displays how many memory modules are installed.
  ***Example:*** *22 modules*
- **Module Details:** Lists form factors, part numbers, and sizes.
  ***Example:*** *M321RYGA0B80-CQKZJ – 96 GB each*

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## 4.6 MiPhi aiDAPTIV+ Cache Information

- **Cache Unit Identification:** Identifies each aiDAPTIV+ Cache unit by name (Cache 1 – Cache 4).
- **Model & Specifications:** Provides device model numbers, capacity, and firmware version.
  ***Example:*** *PASCARI A1802K031T00E004T09000 – 1.02 TB – Firmware: EIFZ61.5*
- **Serial Numbers:** Enables tracking and service reference.

## 4.7 Storage Utilization Summary

- **Boot Drive Status:** Graphically and numerically shows how much storage is used vs. available.
  ***Example:*** *866 GB of 1.8 TB used*
- **Visual Indicators:** Color-coded bar (e.g., blue/red) represents usage health.
  - 🔵 **Blue**: Usage is **below 90%** – healthy range
  - 🔴 **Red**: Usage is **90% or above** – critical threshold; action recommended

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# 5. DATASETS

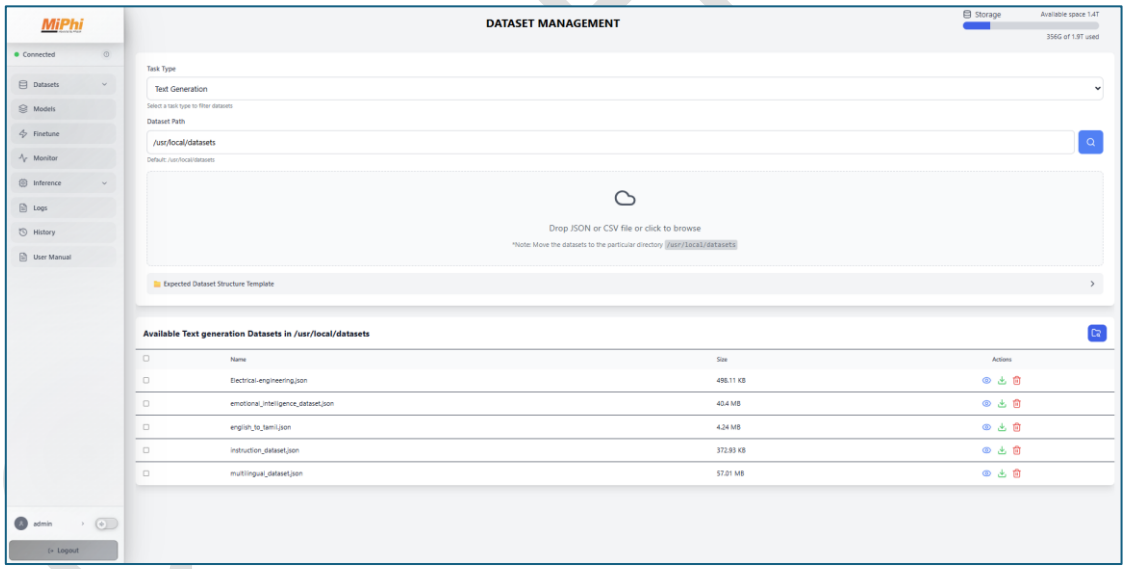
## 5.1 Dataset Management

The **MiPhi Datasets** module provides a unified and task-aware interface for managing training datasets.

The dataset page is organized around **task type selection**, where each task dynamically controls:

- Dataset storage path
- Allowed file or folder format
- Upload validation rules
- Dataset listing and preview behavior

Once a task type is selected, the upload area and dataset list automatically update to reflect the requirements of that task.




### 5.1.1 Task Type Selection

The following task types are supported:

- **Text Generation**
- **Image Text-to-Text**
- **Automatic Speech Recognition (ASR)**
- **Zero-shot Image Classification**

Selecting a task type determines the dataset structure, supported formats, and storage location.

## 5.2 Dataset Handling By Task Type

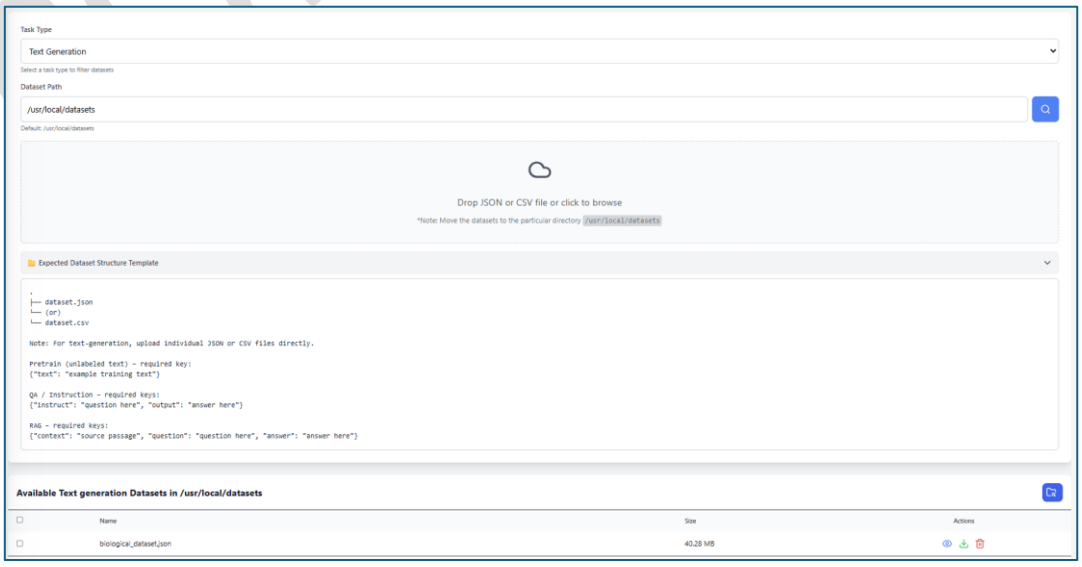
### 5.2.1 Text Generation

**Upload Functionality**

- Supported formats: **.json**, **.csv**
- Users must select a dataset storage path before uploading.
- A **dataset template** is displayed below the upload area for reference.
- Uploaded files are copied to the selected dataset path.

**Dataset Listing**

- Displays all **.json** and **.csv** files in the selected path.
- Each dataset entry shows:
    - File name and size.
- Available actions:
    - 👁️ **Preview** – view dataset content in a modal
    - ⬇️ **Download** – download the file locally
    - 🗑️ **Delete** – remove the dataset with confirmation




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### 5.2.2 Image Text-to-Text

**Upload Functionality**

- Supported format: ZIP folder
- The ZIP must contain:
    - An images/ directory
    - A corresponding .json annotation file
- Folder structure is validated before upload.
- On successful validation:
    - The ZIP file is automatically extracted.
    - Contents are stored in the selected dataset path.
- A folder structure template is displayed below the upload area.

**Dataset Listing**

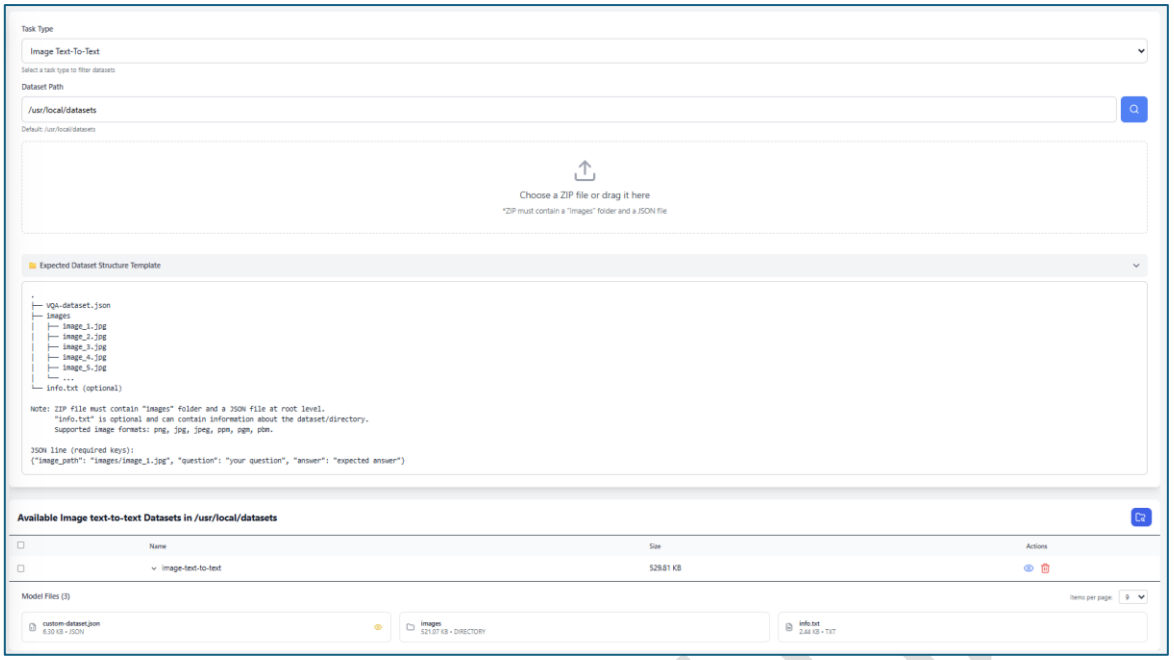
- Displays extracted dataset folders.
- Each folder contains:
    - images/ directory
    - .json annotation file

For each dataset folder, users can:

- View folder name and total size
- 🗑️ Delete the entire dataset folder
- Expand the folder to view its contents
- 👁️ Preview the JSON annotation file directly from the expanded view

**Note:** Image format supported png, jpg, jpeg, ppm, pgm, pbm.

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### 5.2.3 Automatic Speech Recognition

**Upload Functionality**

- Supported format: ZIP folder
- The ZIP must contain:
    - An audio_files/ directory
    - A corresponding .json annotation file
- Folder structure is validated before upload.
- On successful validation:
    - The ZIP file is automatically extracted.
    - Contents are stored in the selected dataset path.
- A folder structure template is displayed below the upload area.

**Dataset Listing**

- Displays extracted dataset folders.
- Each folder contains:
    - audio_files/ directory

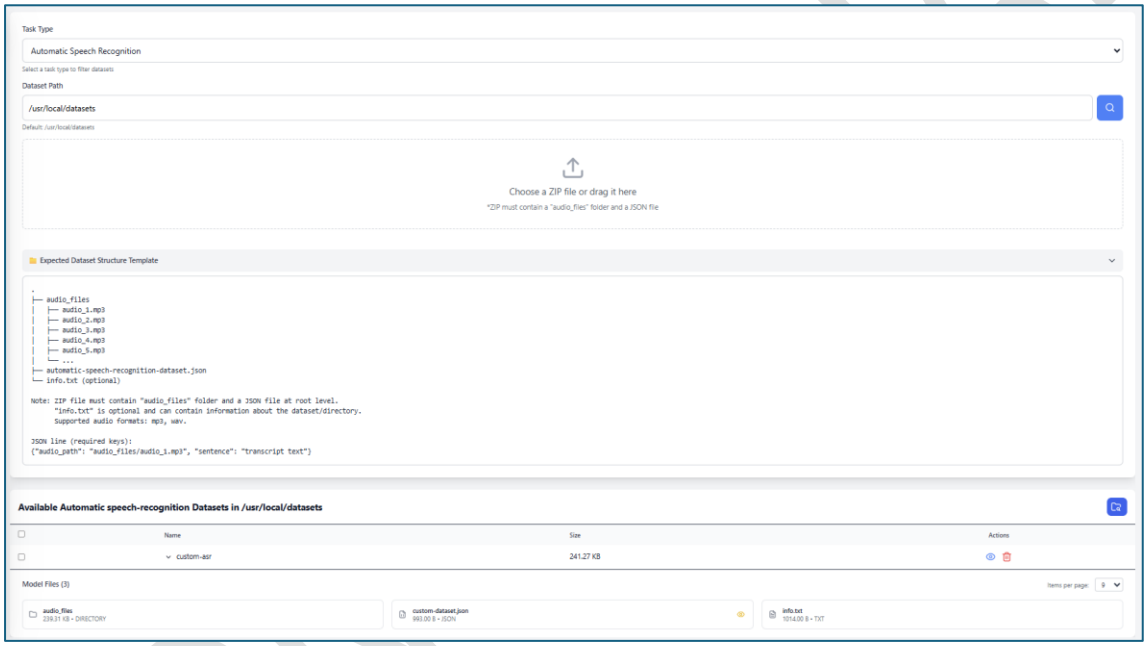
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- .json annotation file

For each dataset folder, users can:

- View folder name and total size
- 🗑️ Delete the entire dataset folder
- Expand the folder to view its contents
- 👁️ Preview the JSON annotation file directly from the expanded view

**Note:** Audio format supported mp3, wav.




### 5.2.4 Zero-shot Image Classification

**Upload Functionality**

- Supported format: ZIP folder
- The ZIP must contain:
  - An images/ directory
  - A corresponding .json annotation file
- Folder structure is validated before upload.
- On successful validation:

- The ZIP file is automatically extracted.
- Contents are stored in the selected dataset path.

- A folder structure template is displayed below the upload area.

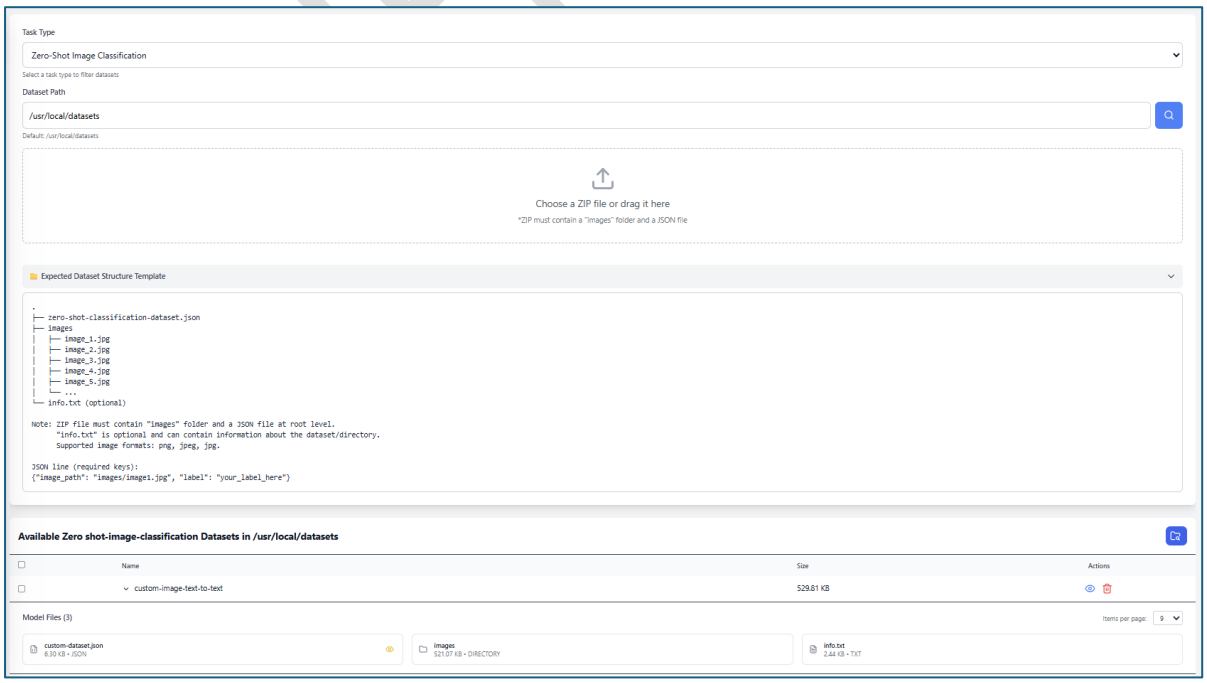
**Dataset Listing**

- Displays extracted dataset folders.
- Each folder contains:
  - images/ directory
  - .json annotation file

For each dataset folder, users can:

- View folder name and total size
- 🗑️ Delete the entire dataset folder
- Expand the folder to view its contents
- 👁️ Preview the JSON annotation file directly from the expanded view

**Note:** Image format supported png, jpg, jpeg.



## 5.3 Features Available For All Dataset Types

**Storage Usage**

- Displays total used space and remaining available storage.
- Automatically updates after uploads or deletions.

**Validation & Error Handling**

- Invalid formats or incorrect folder structures are rejected.
- Clear validation messages guide users to correct dataset structure.
- Duplicate files trigger Replace / Skip options.

**Search**

- Search box allows filtering datasets by name.
- Supports partial and full matches.
- Results update in real time.

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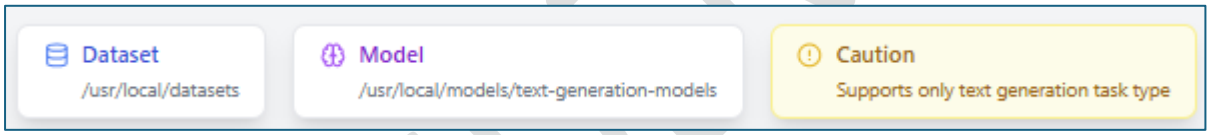
# 6. DATASET GENERATOR

The **MiPhi Dataset Generator** simplifies the process of converting PDF, DOC, DOCX, TXT, HTML, PPTX documents into structured datasets for fine-tuning machine learning models. Designed with an intuitive interface, this module lets users specify the output location, upload a PDF or DOC or DOCX or TXT or HTML or PPTX file, name the output dataset, and generate it with a single click. Real-time progress tracking and a preview option ensure transparency and accuracy throughout the process.

## 6.1 Output Directory

This field automatically uses the directory paths specified in the Datasets page and Models page for saving the generated dataset.

- The generator saves the dataset to the predefined locations configured in the respective pages upon successful generation.

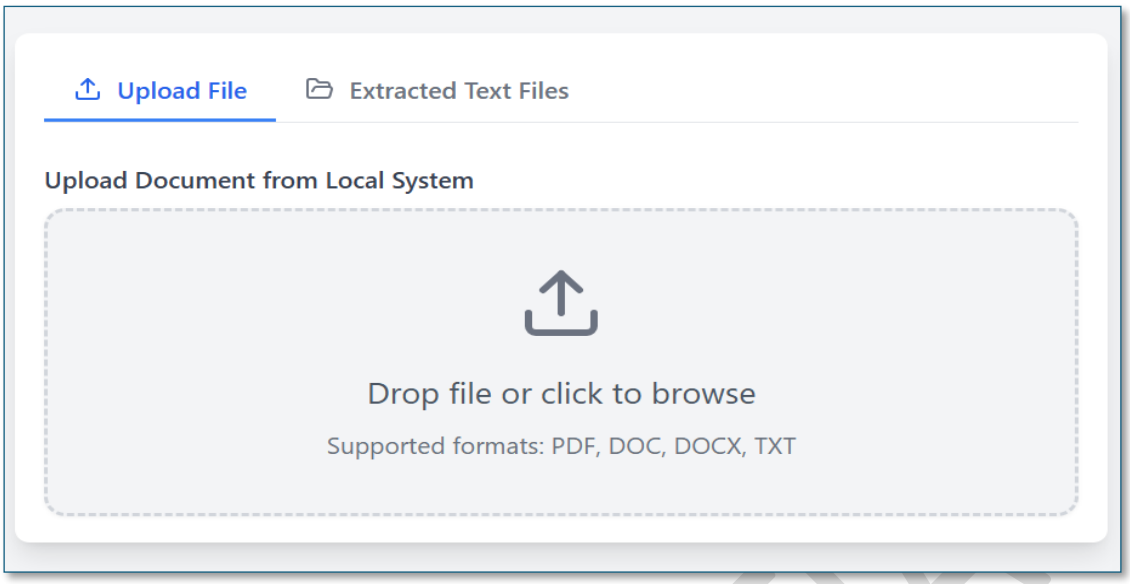



**Note:** The Dataset Generator supports only the **Text Generation** task type and can generate datasets in PreTrain, QA, and RAG formats.

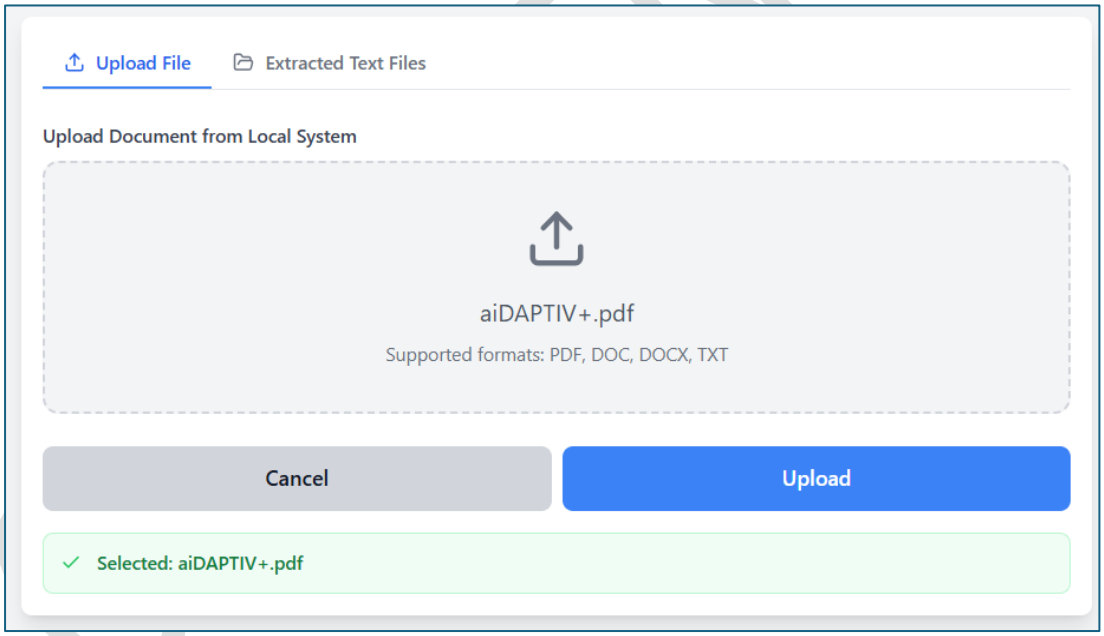
## 6.2 Upload File

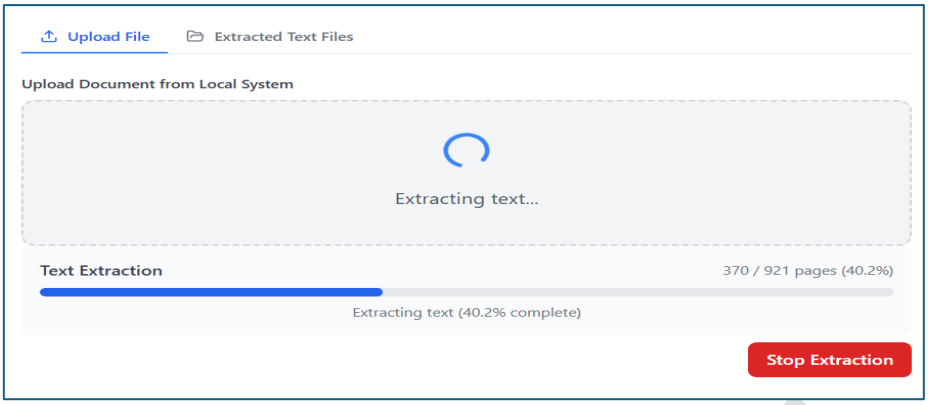
- **Upload File Tab:**
  Upload a document from your local system to begin the workflow:
  **Upload → Text Extraction → Generation**
    - Supported formats: **PDF, DOC, DOCX, TXT, HTML, PPTX**
    - Drag and drop the file or click to browse and upload manually.

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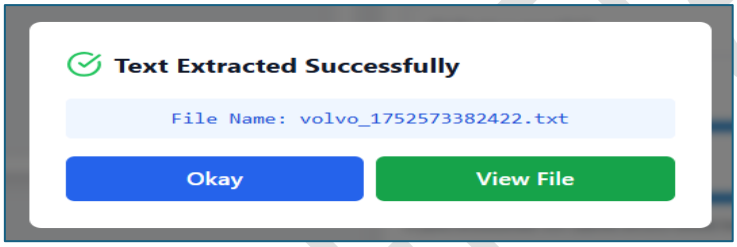



- Once you click upload , the document will be automatically processed for text extraction.



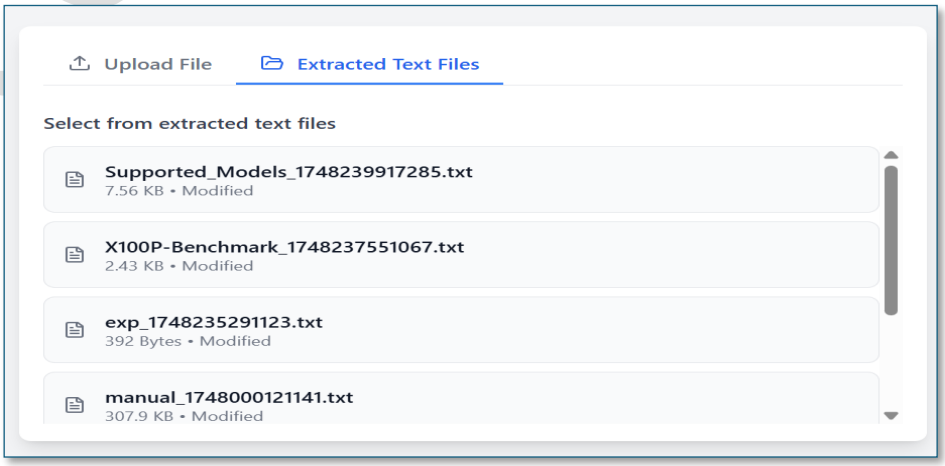



If required, you can stop the extraction in between by clicking the **"Stop Extraction"** button.




Once text extraction is successfully completed, a confirmation message is displayed with the extracted file name. Click **"View File"** to preview the file content.

- **Extracted Text Files Tab:**
  Select from previously extracted text files to continue with:
  **Text Extraction → Generation**
  - Each file is listed with its name, size, and last modified timestamp for verification.




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### 6.3 Generation Parameters

**Dataset File Name**

- Enter the desired name for your dataset file.
- The name entered will be saved with a **.json** extension.
- Special characters are restricted to ensure compatibility with file systems.

**Dataset Type**

- Select the type of dataset to be generated (e.g., QA , Pretrained , RAG ).
- The chosen type determines the structure and formatting of the generated data.

**Number of Pairs**

- Specify how many input-output pairs you want to generate.
- Must be a positive integer value.
- Larger numbers may take longer to generate.

**Select Model**

- Choose the model that will be used for generating the dataset.

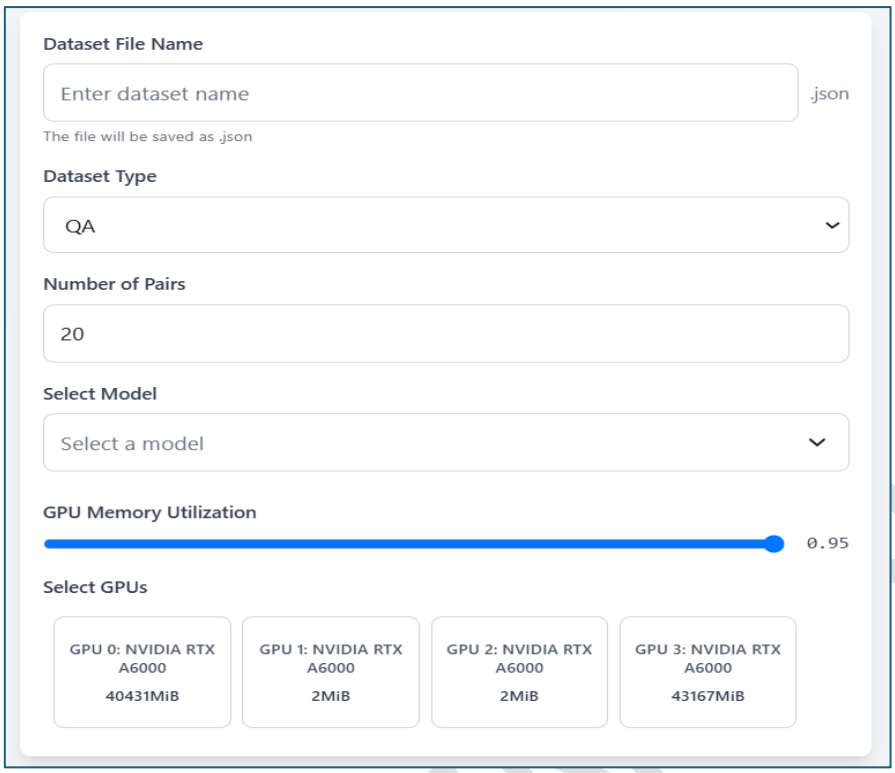
**GPU Memory Utilization**

- Adjust the slider to allocate the percentage of GPU memory for this process.
- Setting this value too high may lead to out-of-memory errors depending on your GPU capacity.

**Select GPUs**

- Select at least **one GPU** from the available list before starting dataset generation.
- This is mandatory to specify which GPU(s) will be used for the process.
- Each GPU card displays its **index, model name, and currently available memory** for your reference.

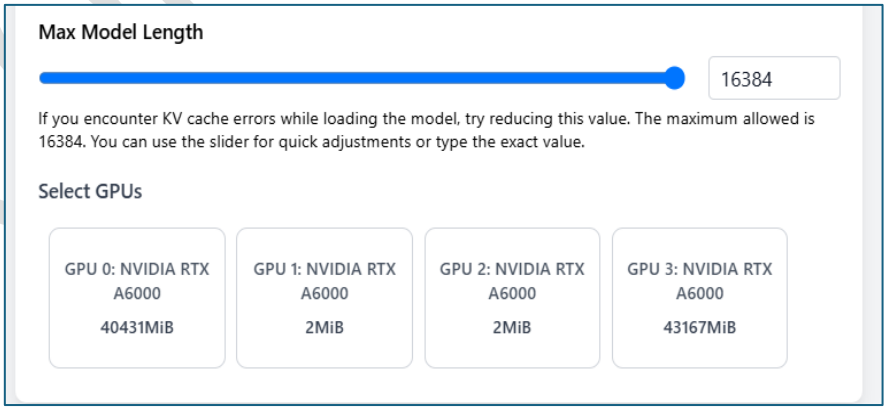
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## 6.4 Generate Dataset

**Provide Required Inputs**

- Enter the **dataset file name** in the input field (e.g. test.json).
- Upload a document file in **PDF, DOCX, DOC, or TXT** format from your local system.
- Select the desired **model** from the dropdown list.
- Adjust the **GPU memory utilization** and **max model length** sliders as needed.




**Note:** The maximum model length option becomes available after selecting a model from the dropdown.

- Select at least **one GPU** from the available list to run the dataset generation.

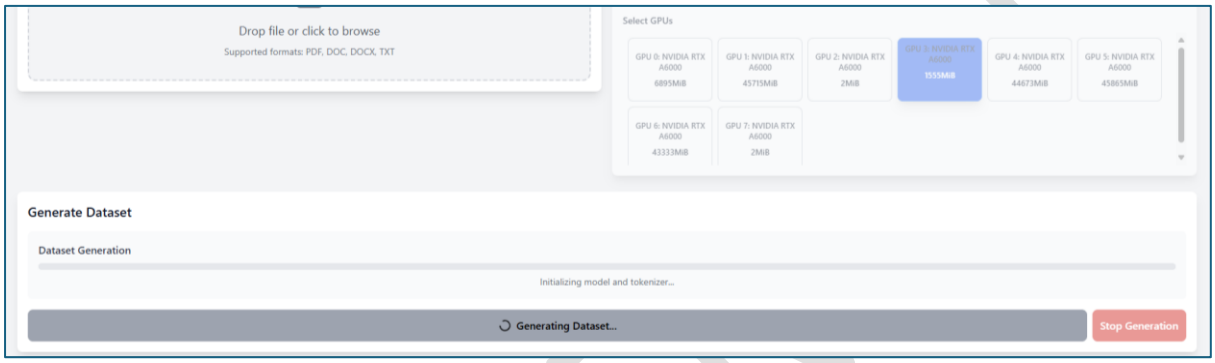
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**Start Dataset Generation**

- Click the **"Generate Dataset"** button to initiate the process.

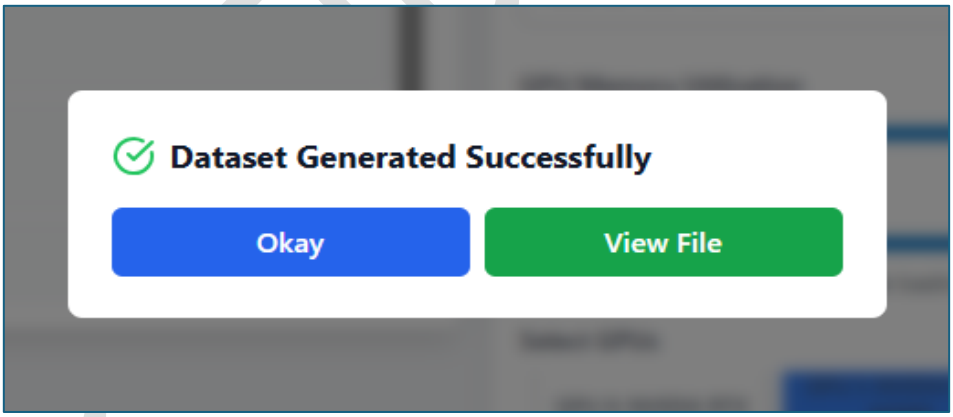
**Monitor Progress**

- A progress bar and status message will appear at the bottom, showing the real-time generation progress (e.g. "Initializing model and tokenizer…" or "Generating Dataset…").
- You can **stop the generation** at any time by clicking the **"Stop Generation"** button if needed.




**Completion**

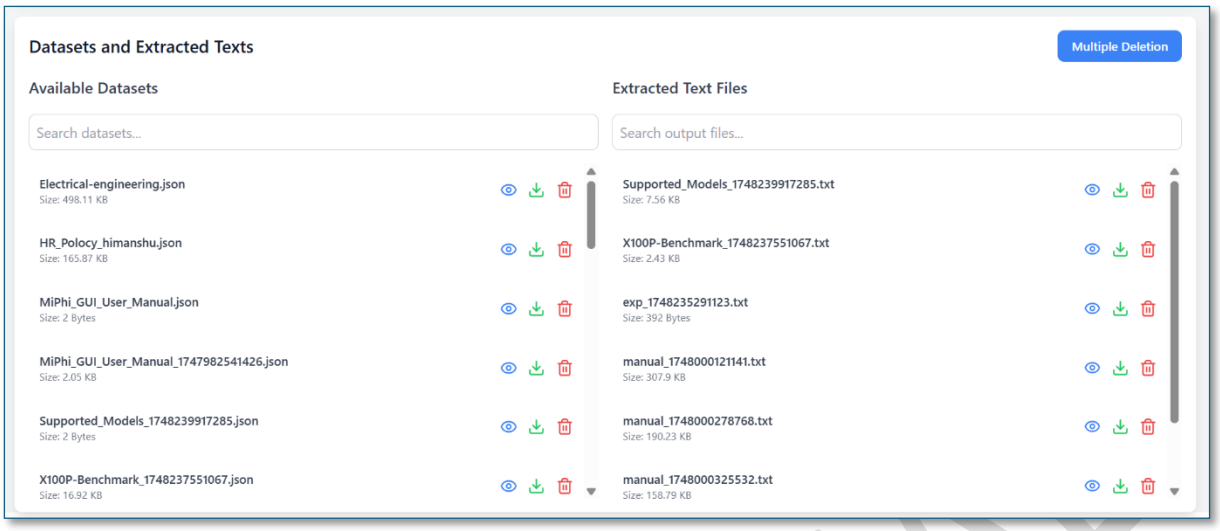
- Upon successful completion, a confirmation pop-up will appear stating: **"Dataset generated successfully."**

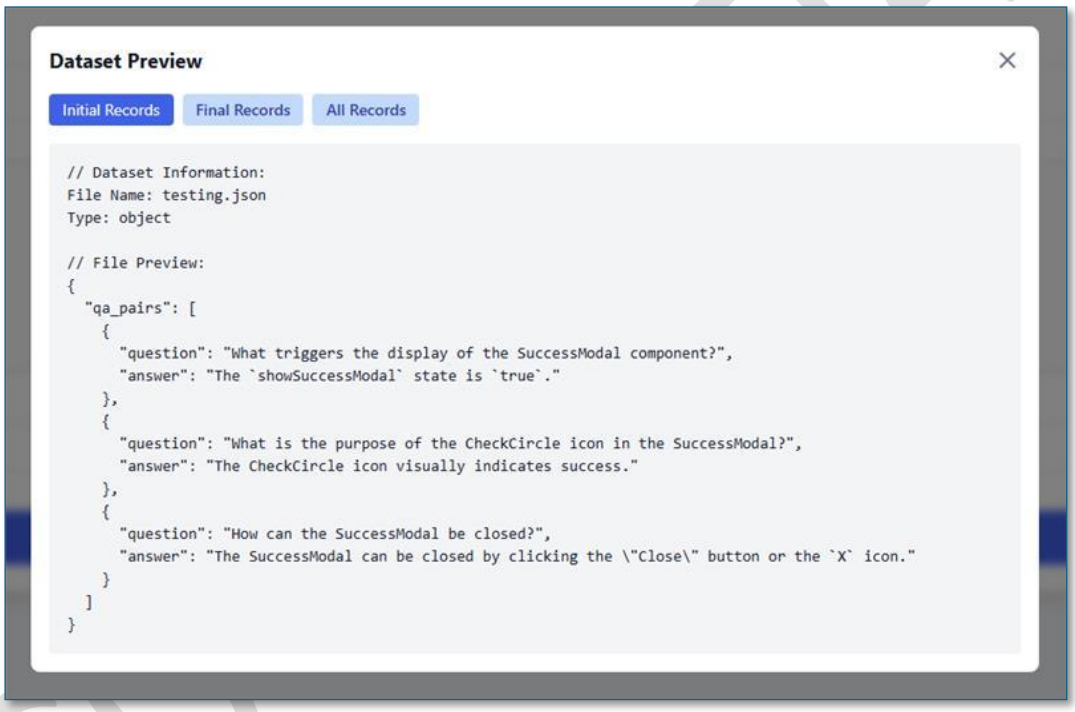



## 6.5 Preview Generated Dataset and Extracted Text files

After successful dataset generation:

- Users can click the **Preview Dataset** button to open and view the structured output.
- This helps validate the extracted content and formatting before using it for training.
- Users can view the list of datasets generated and Extracted text files.

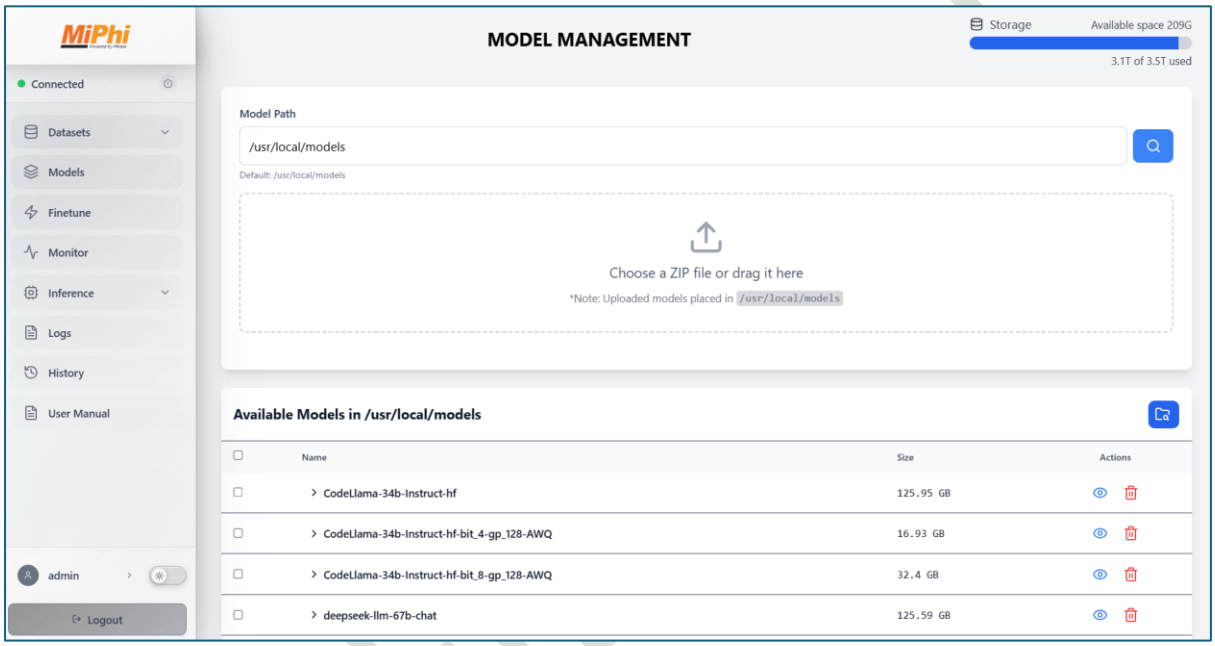

Datasets and Extracted Texts
Multiple Deletion
Available Datasets
Extracted Text Files
Search datasets...
Search output files...
Electrical-engineering.json
Size: 498.11 KB
HR_Polocy_himanshu.json
Size: 165.87 KB
MiPhi_GUI_User_Manual.json
Size: 2 Bytes
MiPhi_GUI_User_Manual_1747982541426.json
Size: 2.05 KB
Supported_Models_1748239917285.json
Size: 2 Bytes
X100P-Benchmark_1748237551067.json
Size: 16.92 KB
Supported_Models_1748239917285.txt
Size: 7.56 KB
X100P-Benchmark_1748237551067.txt
Size: 2.43 KB
exp_1748235291123.txt
Size: 392 Bytes
manual_1748000121141.txt
Size: 307.9 KB
manual_1748000278768.txt
Size: 190.23 KB
manual_1748000325532.txt
Size: 158.79 KB


Dataset Preview
Initial Records
Final Records
All Records
// Dataset Information:
File Name: testing.json
Type: object
// File Preview:
{
"qa_pairs": [
{
"question": "What triggers the display of the SuccessModal component?",
"answer": "The `showSuccessModal` state is `true`."
},
{
"question": "What is the purpose of the CheckCircle icon in the SuccessModal?",
"answer": "The CheckCircle icon visually indicates success."
},
{
"question": "How can the SuccessModal be closed?",
"answer": "The SuccessModal can be closed by clicking the \"Close\" button or the `X` icon."
}
]
}

# 7. MODELS

The **MiPhi Models** is designed to help users manage Large Language Models (LLMs) efficiently through an interactive and organized interface. This page supports operations like loading models from a specific directory, uploading new models, viewing configuration files, and deleting unwanted models.

Model management page consists of two main sections upload area and list area.

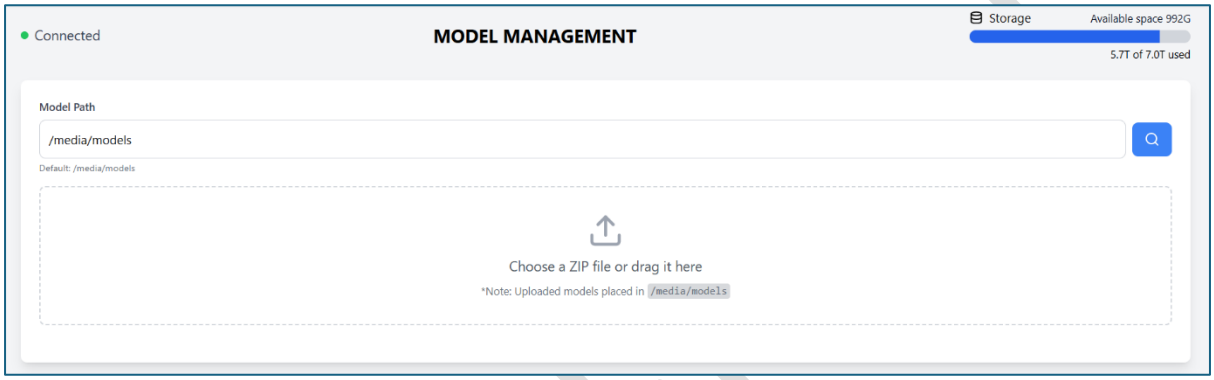



## 7.1 Upload Area

- This section allows you to upload model in **ZIP format only**.
- You can either drag and drop a ZIP file or click to browse and select a file.
- The **default model's storage path** is /media/models, but you can change it as needed.
- If a model with the same name already exists: The system prompts: **"Replace existing model?"**
- Options: Replace | Skip.
- This avoids accidental overwriting and ensures version control.

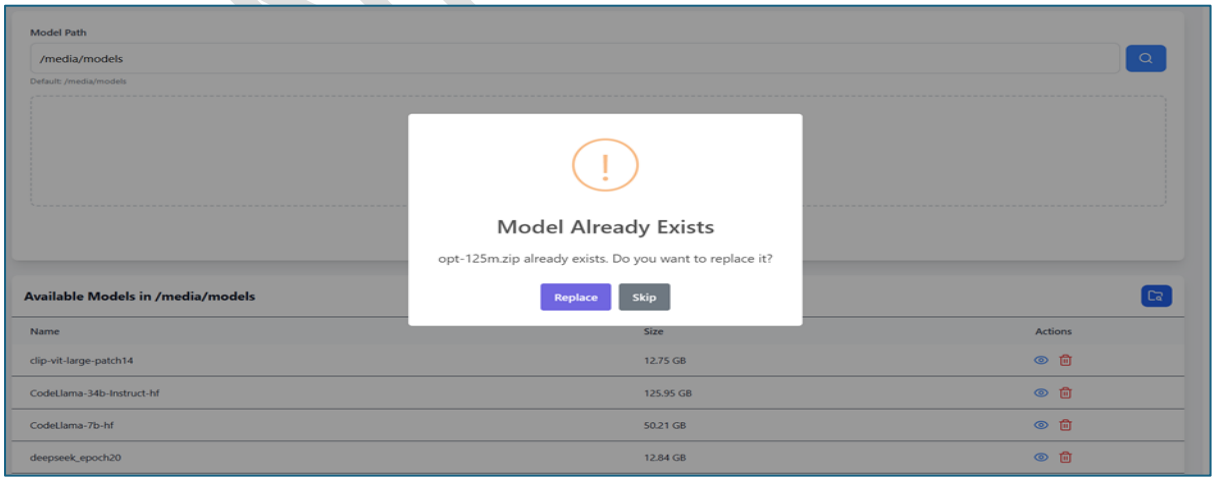
### 7.1.1 Select and Upload Your Model

- Click the Choose a ZIP file or drag it here file to browse area to open your system's file picker.
- In the dialog (e.g. browsing **Downloads**), select your desired model zip file and hit **Open**.
- Once chosen, the file name appears in the upload zone and the dataset is copied into /media/models for immediate use.



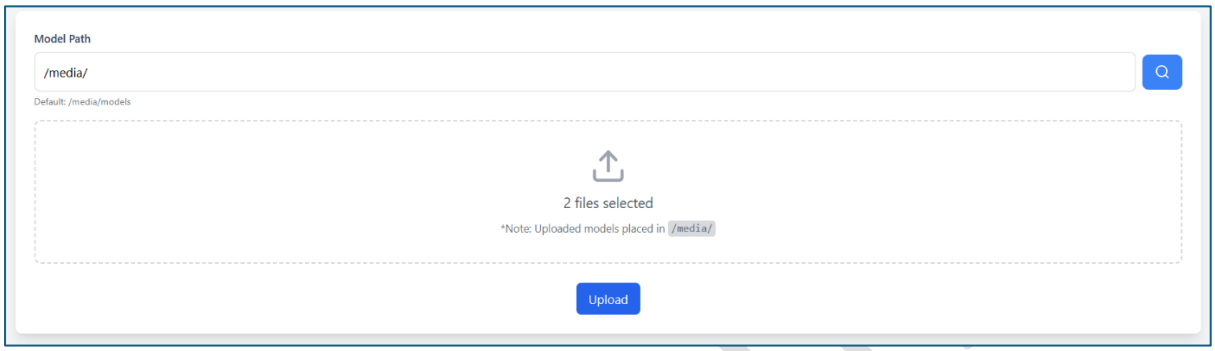

### 7.1.2 File Already Exists

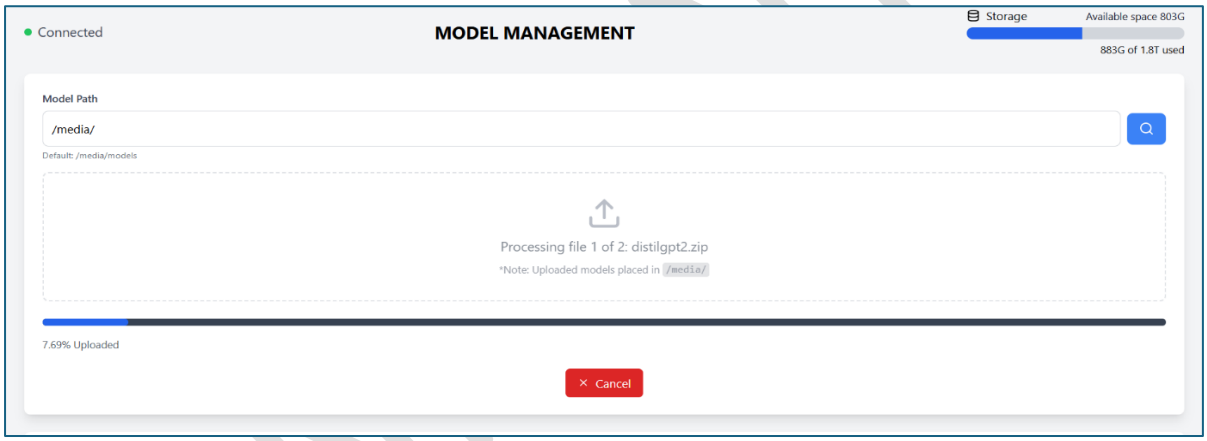
- The system has detected that **opt-125m Model** is already present in /media/models.
- Click **Replace** to overwrite the existing file with this new version.
- Click **Skip** to cancel uploading that file and keep the current copy intact.



### 7.1.3 Upload Multiple Models

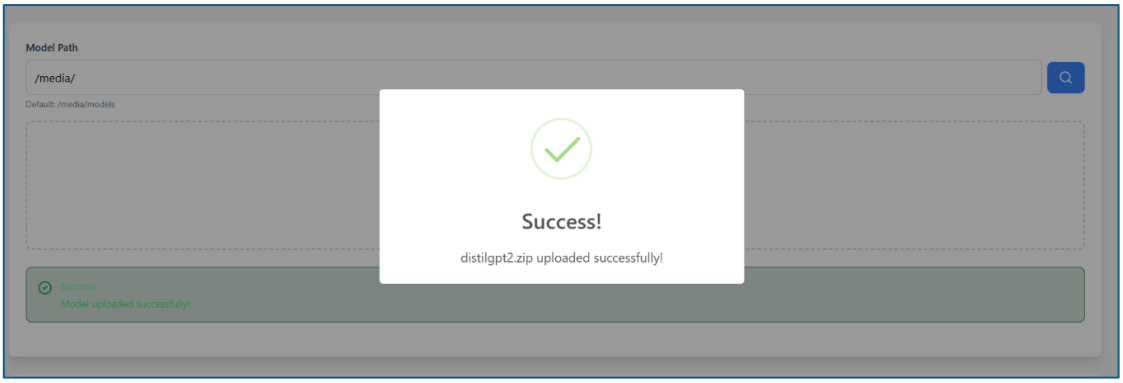
- Click the upload area and, in your file picker, use Ctrl/Cmd + Click or Shift + Click to select several ZIP files at once.
- All chosen files will be listed in the confirmation dialog along with their individual sizes.
- Hit **Upload** to upload the models one after the other.




### 7.1.4 Upload Complete

- The dialog confirms your upload results, showing how many files were successfully uploaded and if any failed.
- Your new models have been added to /media/models and will appear in the “Available Models” list once you click OK.
- Click OK to dismiss and refresh the page, updating storage usage and file listings.



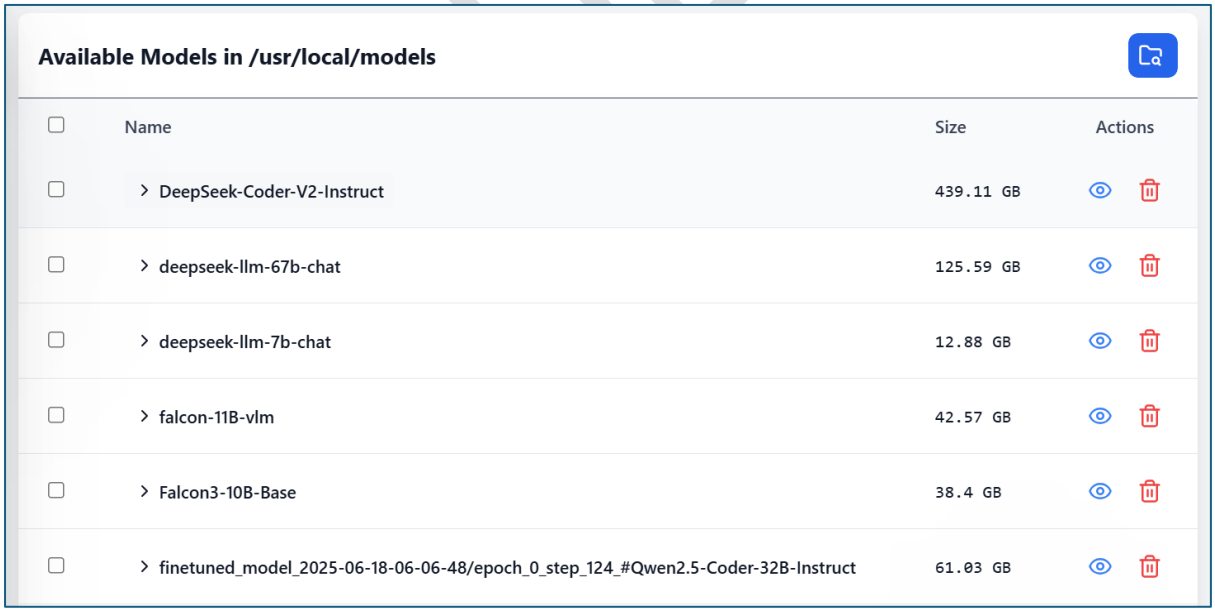

## 7.2 List of Available Models

- Displays all the uploaded models.
- Each dataset entry shows its **name and size**.
- You have the following actions available:

  **View**– To check the model contents.

  **Delete** – To remove the model from storage.

- A **search bar** is provided to quickly find specific models.




### 7.2.1 Model Preview

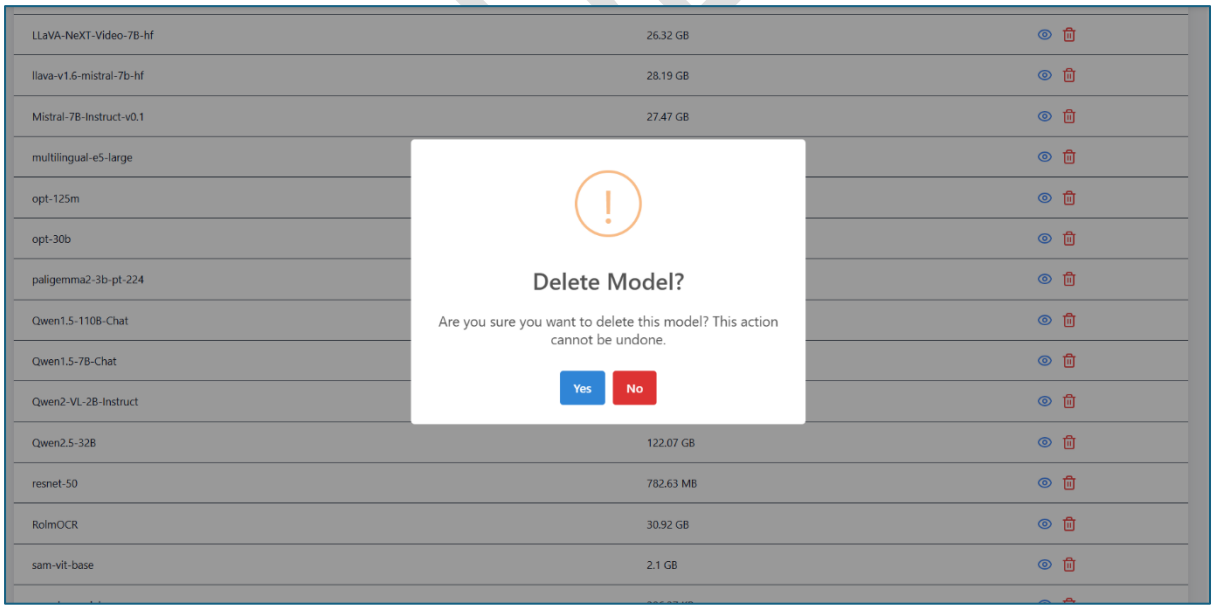
- The configuration details of the model are presented through a scrollable JSON view.
- This allows users to easily inspect the contents of the config.json file for better understanding and verification.

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### 7.2.2 Deleting a Model

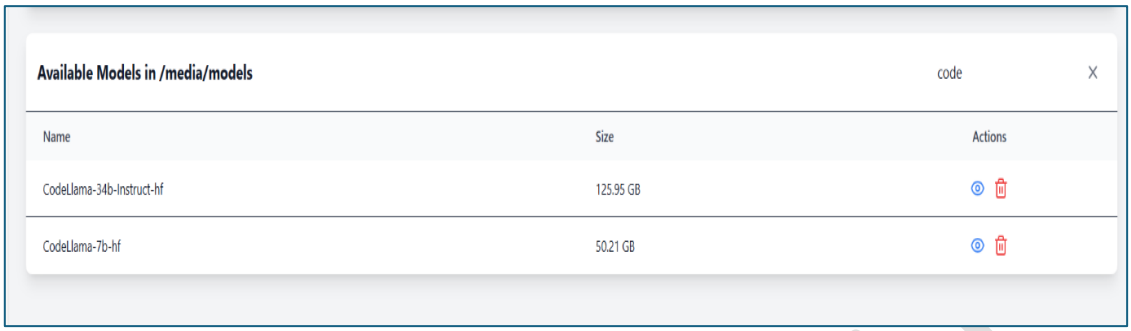
- You're about to permanently remove the selected file from /media/models; this action cannot be undone.
- Click **Delete** to confirm and remove the model immediately.
- Click **Cancel** to abort and keep the file safely stored.




### 7.2.3 Searching a Model

- Use the search box in the top-right of the list to filter files by name or extension—typing "code" instantly shows only matching entries.
- Results update in real time as you type; hit the × icon to clear your query and restore the full list.

- Supports partial and full filename matches (e.g. entering “code” will display only files having the word “code”)




### 7.2.4 Model Explorer – Expand & Navigate Files

**Expand Model to View Contents:**

- Clicking the **dropdown arrow ( ▾ )** next to the model name **expands** the model folder.
- This reveals the files contained within the model directory, such as .py, .json, .safetensors, and .md files.

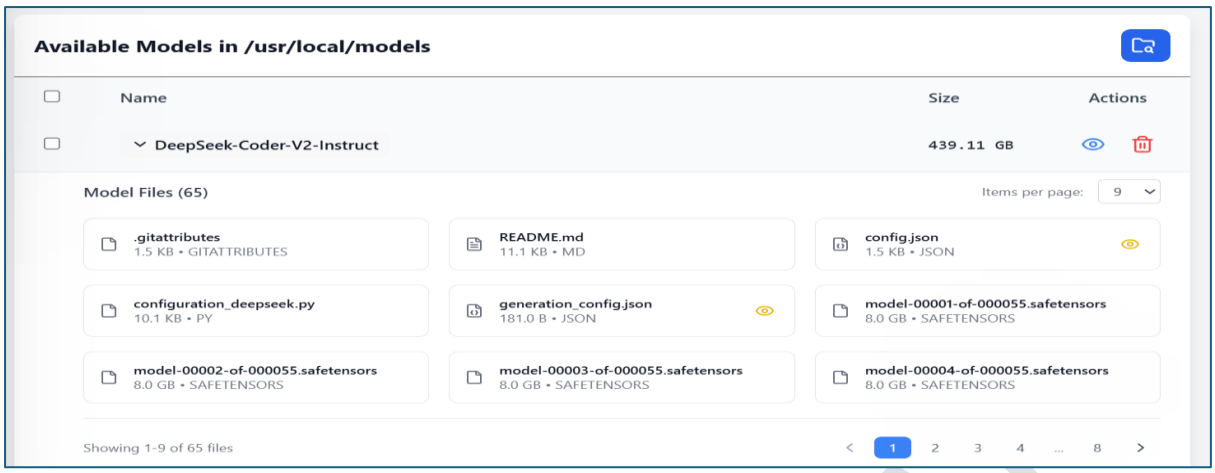
**Pagination of Files:**

- If the model has many files (like 65 in this example), only a **subset of files (default 9)** is shown per page.
- Users can navigate between file pages using the **pagination controls** at the bottom (←, 1, 2, 3, ..., →).
- There's also a **dropdown menu** (Items per page) to adjust how many files are displayed at once.

**File Actions:**

- Each file entry displays the **file name**, **size**, and **type**.
- For viewable files (like JSON or markdown), a **"view" icon ( 👁 )** is available for quick previews.

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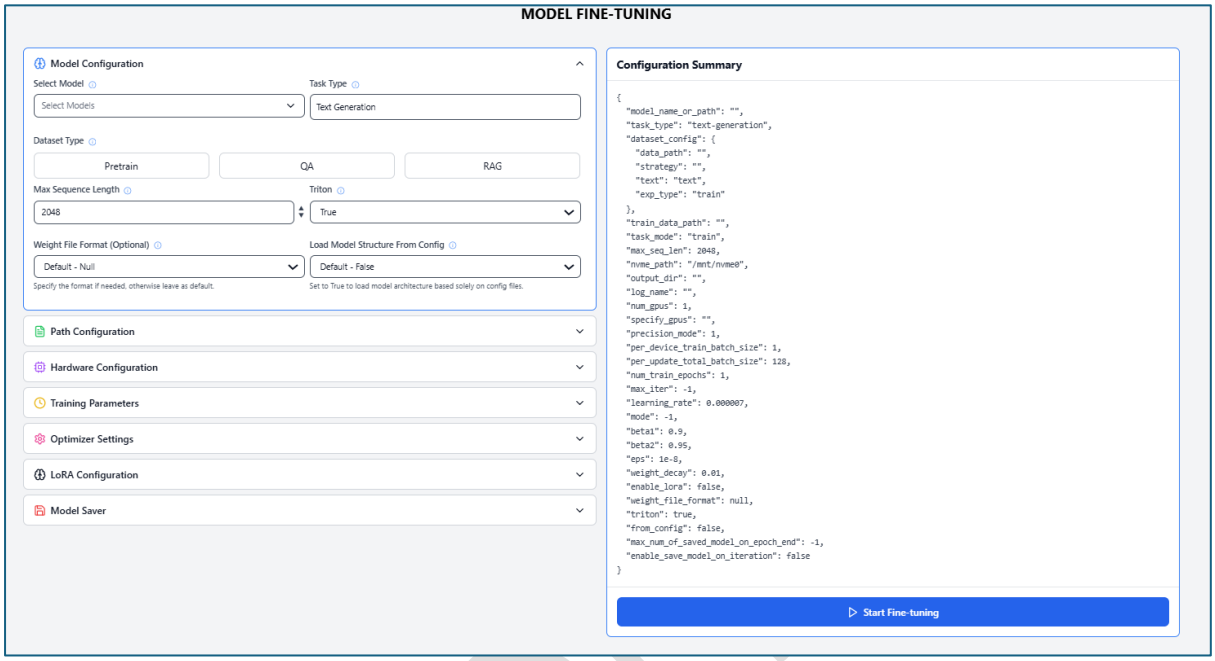




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# 8. FINE-TUNING

The fine-tuning page have two main sections: on left side section assign parameters and on right side section for viewing a configuration summary and then you can start fine-tuning.




## 8.1 Model Configuration Section

This section allows users to set up the parameters for fine-tuning the model. It typically includes:

### 8.1.1 Select Model

- This dropdown lists available models based on the path specified in the Models page.
- When clicked, it displays all models stored in the configured path.

### 8.1.2 Task Type

#### Overview

The **Task Type** defines the learning objective of the model and determines how datasets are interpreted during fine-tuning.

Selecting a task type dynamically updates:

- Supported dataset formats
- Dataset configuration fields
- Required keys and validation rules

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The following task types are supported:

- **Text Generation**
- **Image Text-to-Text**
- **Automatic Speech Recognition**
- **Zero-shot Image Classification**

### **Task Type:** Text Generation

This task is designed for models that generate text based on a prompt.
Typical use cases include chatbots, summarization, storytelling, and code generation.

**Supported Dataset Types**

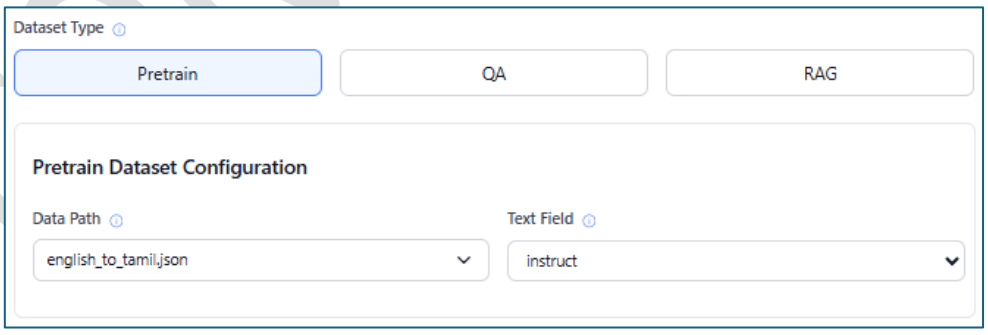
- Pretrain
- QA
- RAG

### 1) Pretrain Dataset

**Purpose:** Train the model on raw text content.

**Dataset Configuration**

- **Data Path:** Select the dataset file from the dropdown list
- **Text Field:** Choose the key containing training text
  (e.g., text, content, body, message)




### 2) QA Dataset

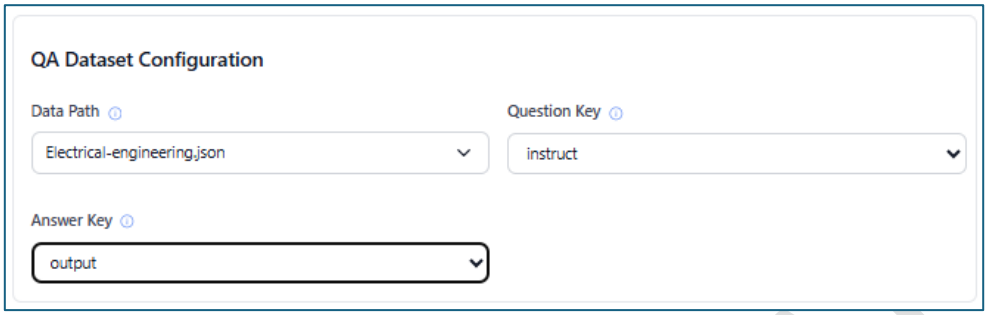
**Purpose:** Train the model to answer questions based on input prompts.

**Dataset Configuration**

- **Data Path:** Select the dataset file from the dropdown list

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- **Question Key:** Select the JSON key representing the question
- **Answer Key:** Select the JSON key representing the expected answer

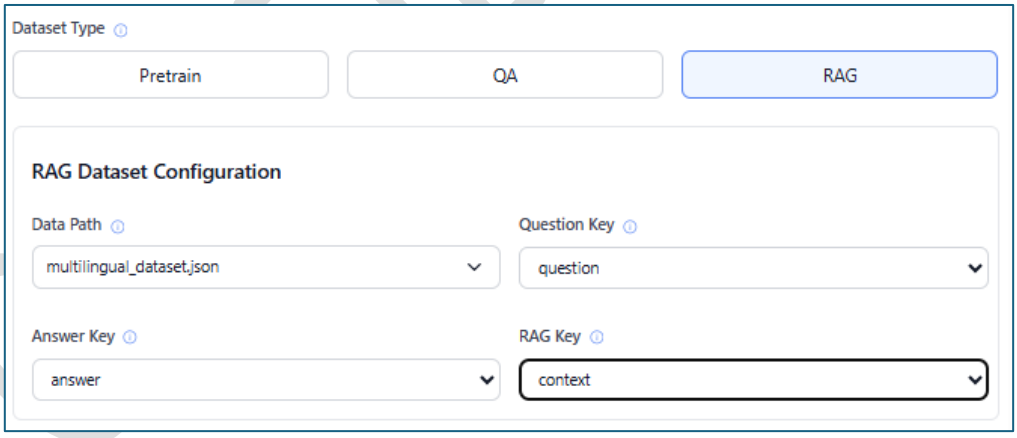



### 3)RAG Dataset

**Purpose:** Enhance generation using retrieved contextual data.

**Dataset Configuration**

- **Data Path:** Select the dataset file from the dropdown list
- **Question Key:** Select the key for the prompt
- **Answer Key:** Select the key for the expected output
- **RAG Key:** Select the key containing retrieval context




### Task Type: Image Text-to-Text

This task supports vision–language learning where images are paired with text.

**Dataset Type**

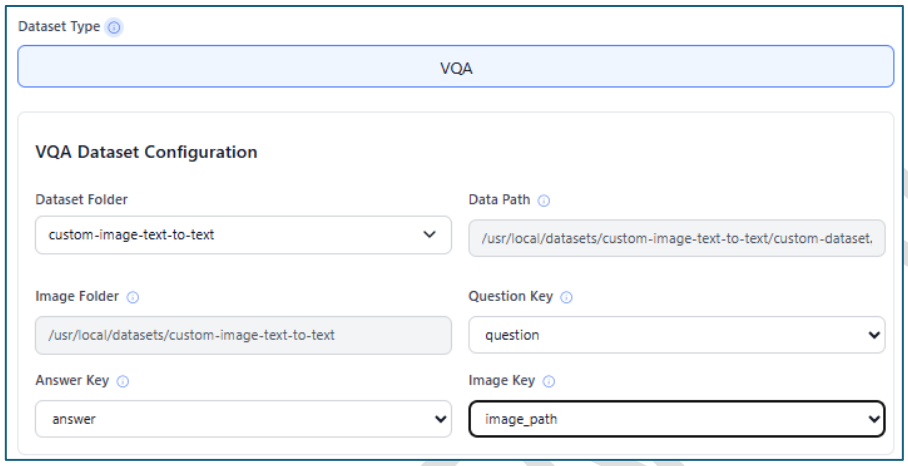
- VQA

**VQA Dataset Configuration**

- **Dataset Folder:** Select the dataset folder which contains images & json file.

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- **Data Path:** Automatically populated with the path of the JSON file inside the folder
- **Image Folder:** Automatically populated with the parent path of the image directory
- **Question Key:** Select from JSON keys
- **Answer Key:** Select from JSON keys
- **Image Key:** Select from JSON keys




## Task Type: Zero-shot Image Classification

This task enables classification without task-specific training labels.

**Dataset Type**

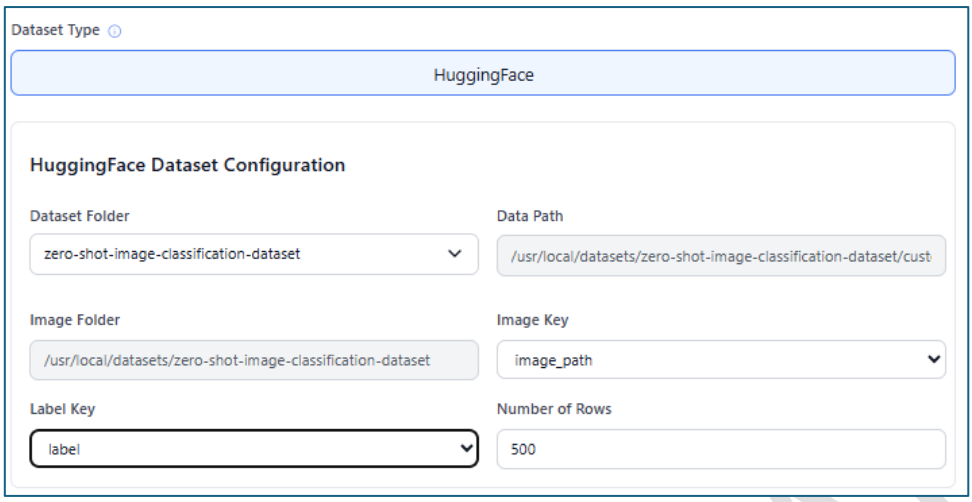
- HuggingFace

**HuggingFace Dataset Configuration**

- **Dataset Folder:** Select the dataset folder which contains images & json file.
- **Data Path:** Automatically populated with the path of the JSON file inside the folder
- **Image Folder:** Automatically populated with the parent path of the image directory
- **Image Key:** Select from JSON keys
- **Label Key:** Select from JSON keys
- **Num Rows:** Specify the number of rows to use from the dataset.
  Ensure the value does not exceed the total number of rows available in the dataset. If a higher value is provided, the system will automatically use all available rows.

⚠️ **Note:** For **Zero-shot Image Classification**, **per-device train batch size must be ≥ 2**. **Triton** and **LoRA** configurations are **disabled** for this task type.

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## Task Type: Automatic Speech Recognition

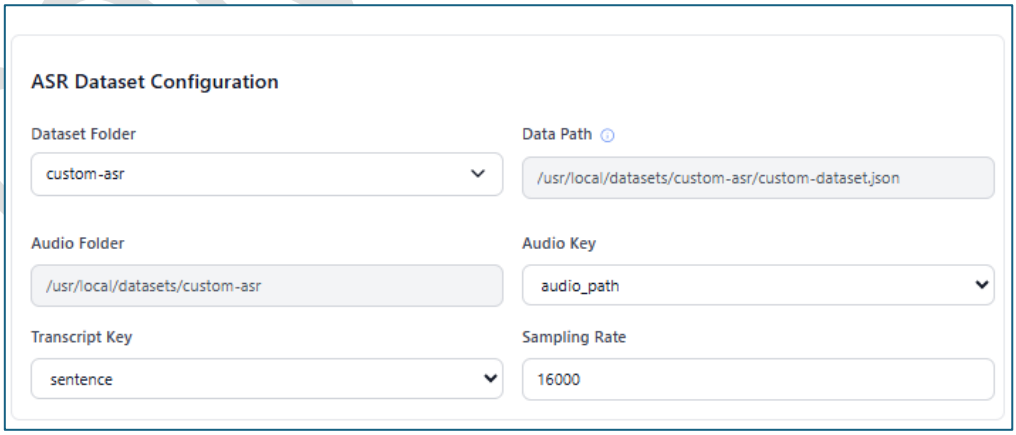
This task is designed for speech-to-text model fine-tuning.

### Dataset Type

- None (Folder-based configuration)

### ASR Dataset Configuration

- **Dataset Folder:** Select the dataset folder which contains audio_files & json file.
- **Data Path:** Automatically populated with the path of the JSON file
- **Audio Folder:** Automatically populated with the parent path of the audio directory
- **Audio Key:** Select from JSON keys
- **Transcript Key:** Select from JSON keys
- **Sampling Rate:** Specify the audio sampling rate (e.g., 16,000 Hz)




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### 8.1.3 Max sequence length

The **max sequence length** defines the maximum number of tokens (words, subwords, or characters) that the model can process in a single input sequence. The supported range depends on the selected model's architecture and configuration.

**Supported Range: 128 to 32768 tokens**

**Default Value: 2048 tokens**

**Recommended Values by Use Case**

- **128 tokens** – Best for very short text inputs such as tweets, simple prompts, and lightweight classification tasks.
- **256 tokens** – Suitable for short paragraphs, emails, and FAQ-style inputs.
- **512 tokens** – Standard for most NLP tasks including document classification, named entity recognition (NER), and summarization.
- **1024 tokens** – Ideal for longer inputs such as news articles, research abstracts, and complex question–answering.
- **2048 tokens** – Commonly used for multi-paragraph text generation, document analysis, and long-form conversations.
- **4096–8192 tokens** – Suitable for long documents and extended conversational context (model-dependent).
- **16384–32768 tokens** – Designed for long-context models handling large documents or multi-section inputs (only supported by specific architectures).

### 8.1.4 Triton

Enables the use of Triton-optimized techniques to improve training efficiency. Currently supports LLaMA, Mistral, and Mixtral models. (Default: True)

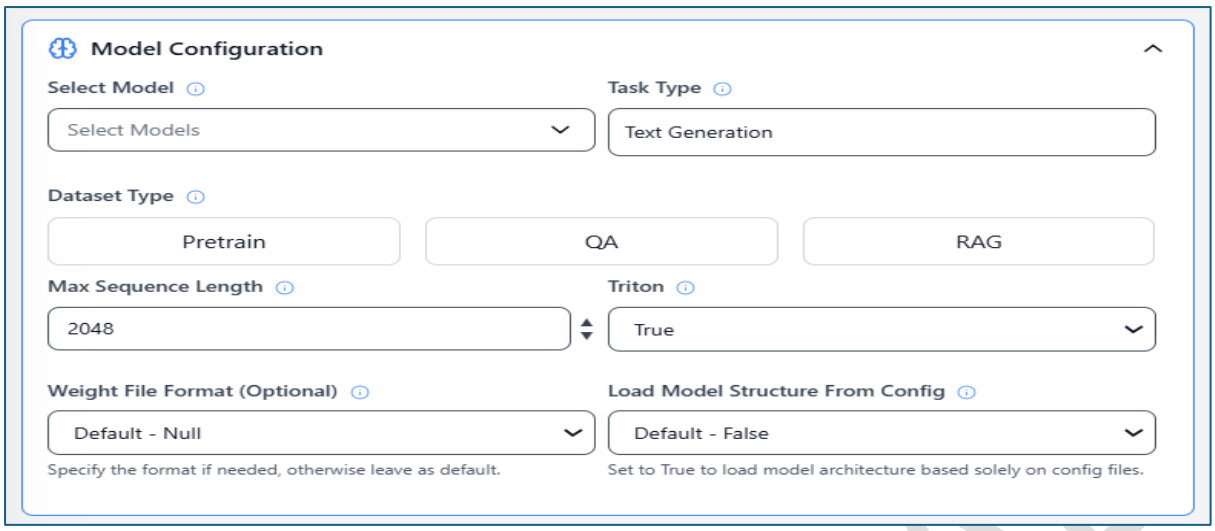
### 8.1.5 Weight File Format

This parameter specifies the format of the model's weight file, such as .bin, .pt, or .safetensors, depending on the type of model being used or required.

### 8.1.6 Load Model Structure From Config:

When enabled, this parameter ensures that the model's architecture is loaded entirely from the config.json file.

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## 8.2 Path Configuration Section

The Path Configuration Section is crucial for defining where various files and logs will be stored during the fine-tuning process. It typically includes:

### 8.2.1 aiDAPTIV+ Cache Mount Point

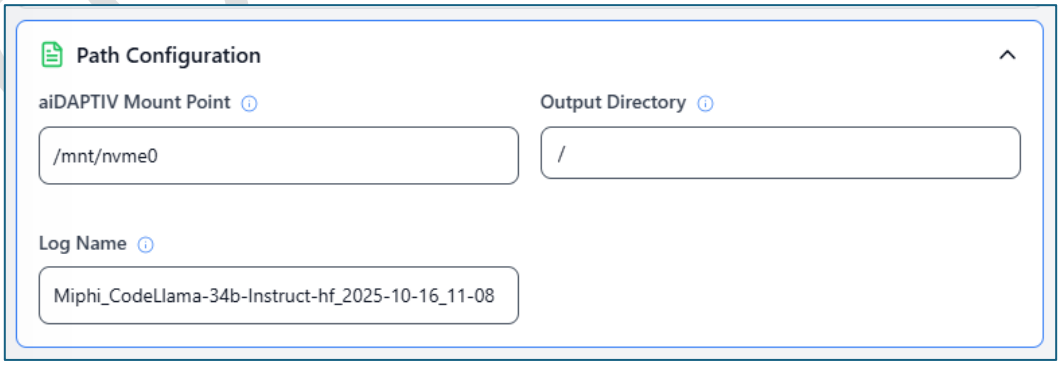
It serves as a high-speed storage location for caching and accessing large amounts of data efficiently. It plays a crucial role in storing and loading model parameters, datasets during training and fine-tuning.

### 8.2.2 Output_dir

Directory where the fine-tuned model will be saved. (But you can change it as needed.).

### 8.2.3 Log_name

The log name is the filename used to track the training process. It will be automatically assigned based on the selected model.




## 8.3 Hardware Configuration Section

This section typically includes options to select GPUs and set the precision mode.

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### 8.3.1 Select GPUs

This option allows you to choose which GPUs will be used for the training process. It also displays the memory usage of each GPU, helping you manage resource allocation efficiently. The available GPU selection may vary based on server availability.

### 8.3.2 Precision Mode

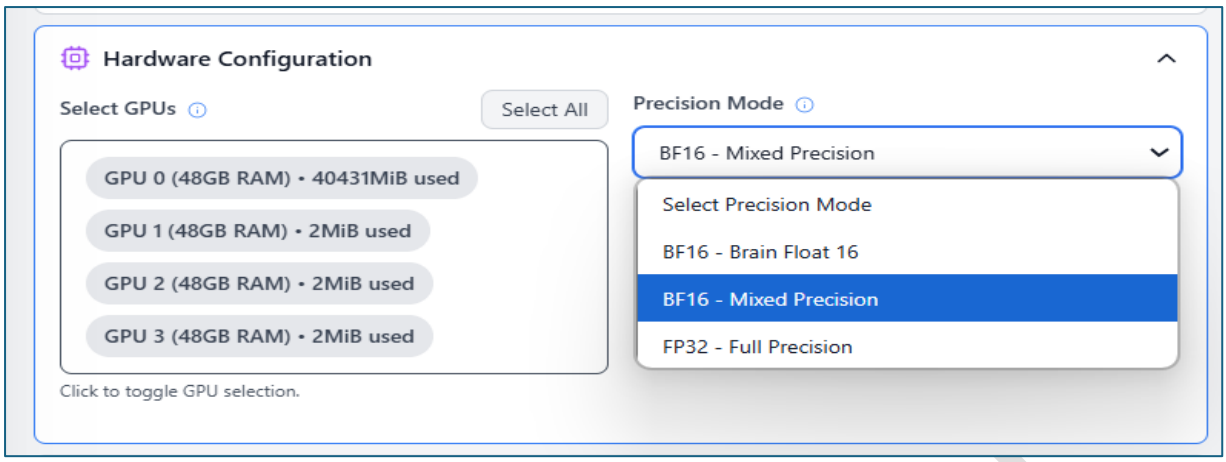
The Precision Mode setting determines how numerical values are processed during model training. It impacts **accuracy, memory usage, and computational speed**.

- **FP32 (Single Precision)**: Standard 32-bit floating point, offering high accuracy but slower computation.
- **FP16 (Half Precision)**: 16-bit floating point, faster than FP32 but with reduced precision.
- **BF16 (Bfloat16)**: 16-bit floating point with a wider exponent range than FP16, balancing speed and accuracy.

You can select from the following precision options

| Precision Mode | Memory Usage | Speed | Accuracy | Best Use Case |
|---|---|---|---|---|
| FP32 | High | Normal | High | High-precision models, sensitive computations |
| FP16 | Low | Fast | Moderate | Deep learning, speed-optimized training |
| BF16 | Moderate | Fast | High | Fine-tuning large models, balanced efficiency |

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## 8.4 Training Parameters Section

The **Training Parameters** section defines key settings that control the training process, including how data is processed in batches, the total number of training steps, and how long the model trains.

### 8.4.1 Train Batch Size

- The number of training samples processed per GPU in one forward and backward pass.
- Affects memory usage and training speed. Larger batch sizes improve training stability but require more GPU memory.

### 8.4.2 Total Batch Size

- The total number of samples used for one update of model weights
- Helps control gradient updates when training across multiple GPUs or accumulating gradients over multiple steps to fit larger batch sizes in memory.

**Example:** If you are running on 4 GPUs with Total batch size =4 and want to update the model every 80 batches, then you should set the total_batch_size to 80.

The machine will run 80/(4 GPUs×4 Batch Size per GPU) =5 iterations and update the model once.

Note: If not divisible, round up to the next whole number.

### 8.4.3 Epoch

- The number of times the model processes the entire dataset during training.
- Controls how long the model learns from the data. Too few epochs might lead to underfitting, while too many can cause overfitting.

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**Example:** If your dataset has **100,000 samples** and you set batch size = 1000, then:

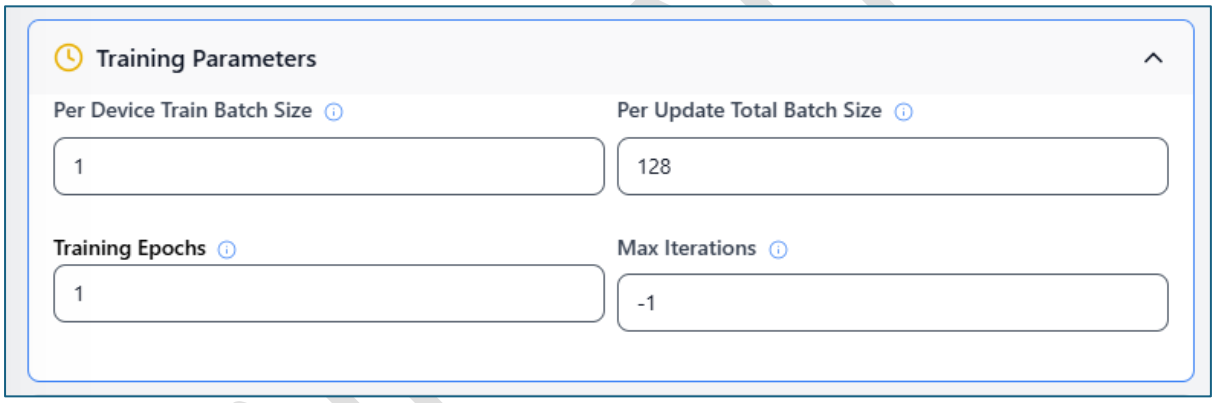
- **1 epoch** = **100 iterations**
- If training for **5 epochs**, the model will perform **500 iterations**.

### 8.4.4 Max Iteration

- The maximum number of training steps (iterations) to perform during training
- Controls when to stop training, independent of epochs.

**Example:**

- If batch size = 32 and dataset has **320,000 samples**, then **1 epoch** = **10,000 iterations**.
- If max_iterations = 20,000, the training will stop after **2 epochs**, even if more epochs are set.




## 8.5 Optimizer Settings Section

The **Optimizer Settings** section defines how the model updates its parameters during the fine-tuning process.

### 8.5.1 Learning rate

- It is the step size at which the model's weights are updated during training. It controls how much the model learns from the gradients at each step.
- A well-chosen learning rate ensures effective training—too high can cause instability, and too low can result in slow convergence.

### 8.5.2 LR Scheduler Mode

- Determines how the learning rate changes over time during training.

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- Adjusting the learning rate during training helps prevent overfitting and improves convergence.
- mode: choose a learning rate mode (default: -1).
- mode == -1: LinearLR (optimizer, start_factor=1, total_iters=1)
- mode == 0: LinearLR (optimizer, start_factor=0.5, total_iters=20) mode == 1: CosineAnnealingLR (optimizer, T_max=150)
- mode == 2: ExponentialLR (optimizer, gamma=0.99)
- mode == 3: MultiplicativeLR (optimizer, lr_lambda=lambda epoch: 0.95)
- mode == 4: StepLR (optimizer, step size=30, gamma=0.1)
- mode == 5: MultiStepLR (optimizer, milestones= [30,80], gamma=0.1)

### 8.5.3 Beta1

Adam Optimizer Hyperparameter(default:0.9)

- The exponential decay rate for the first moment estimate (momentum term) in the Adam optimizer.
- Controls how much past gradients influence the current update, helping smooth training.
- Range: Typically, between 0.8 and 0.99

### 8.5.4 Beta2

Adam Optimizer Hyperparameter(default:0.95)

- The exponential decay rate for the second moment estimate (variance) in the Adam optimizer.
- Helps stabilize the optimization process by adjusting for variance in gradients.
- Range: Typically, between 0.95 and 0.9999

### 8.5.5 Epsilon

hyper-parameter for adam optimizer (default:1e-8).

- A small constant added to prevent division by zero in the Adam optimizer.

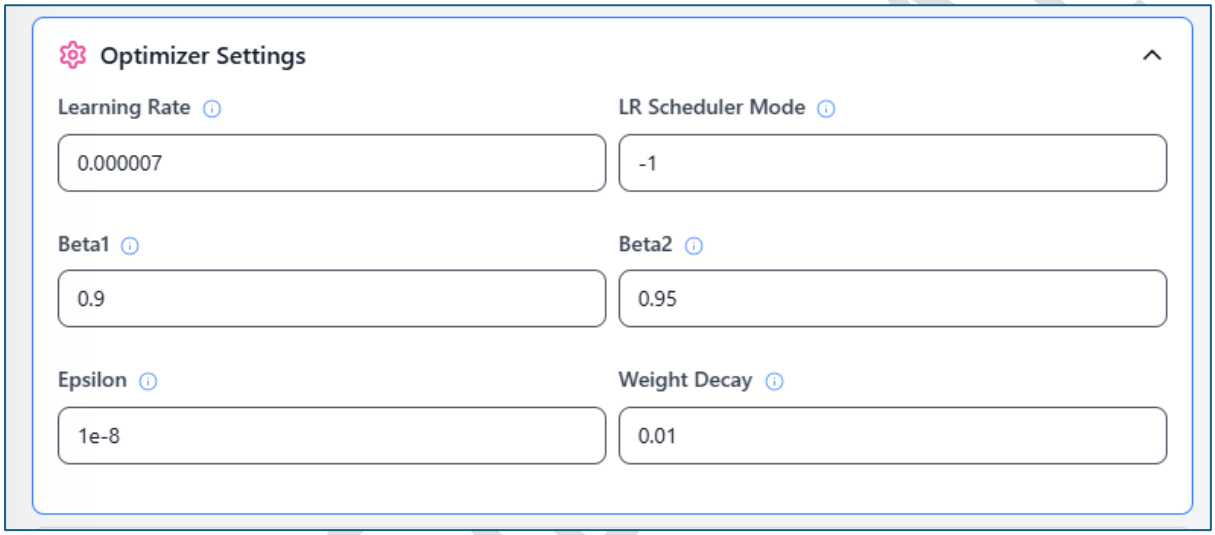
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- Helps maintain numerical stability when computing updates.
- Range: 1e-8 to 1e-6 (Lower values are generally stable.)

### 8.5.6 Weight decay

Set weight decay coefficient (default:0.01)

- A regularization parameter that applies L2 penalty to model weights to prevent overfitting.
- Helps reduce the complexity of the model by discouraging large weights.




## 8.6 LoRA Configuration Section

All LoRA configuration fields will be enabled only when **Enable LoRA** is checked.

### 8.6.1 Enable LoRa

Trigger lora training procedure (default: disabled).

### 8.6.2 LoRA Weight

Absolute path to the pretrained LoRA weight folder.

### 8.6.3 LoRA Optimizer

Specifies the optimizer used during LoRA fine-tuning (e.g., AdamW, Adafactor).

### 8.6.4 LoRA Output Directory

Absolute path for saving the LoRA model.

(Default: None — the LoRA model will only be saved if you provide lora_output_dir.)

### 8.6.5 LoRa rank

Dimension of the low-rank matrices for lora (default: 8).

### 8.6.6 LoRa alpha

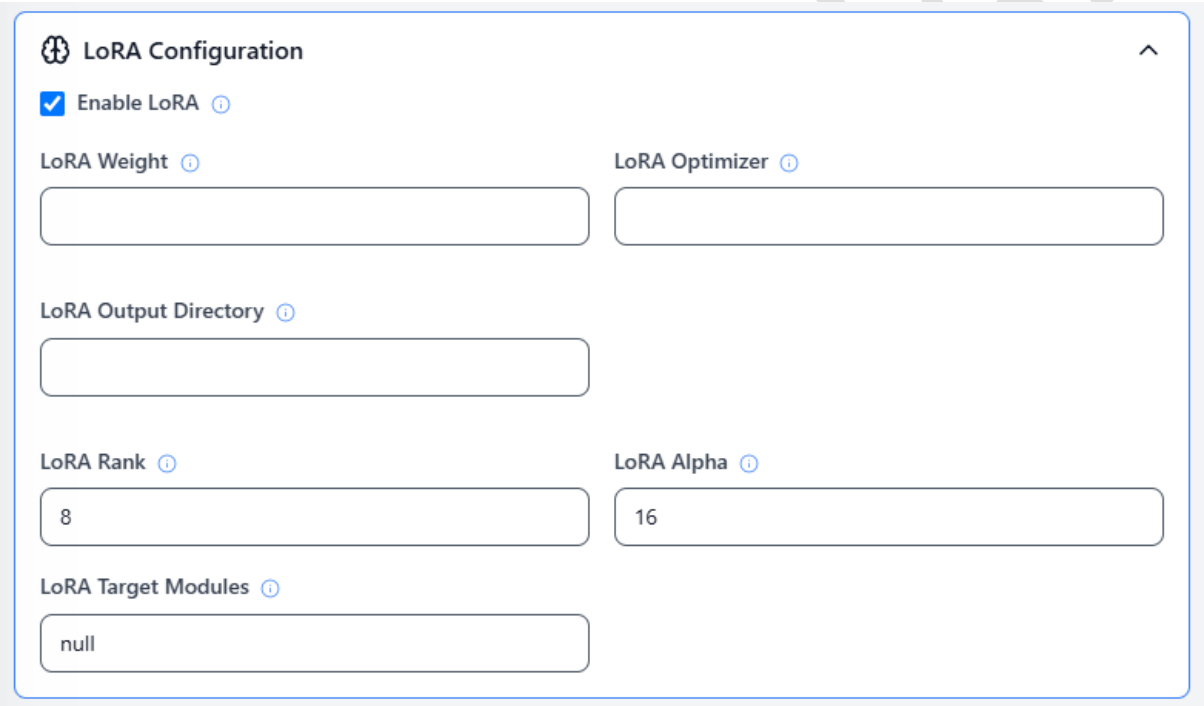
LoRA Alpha adjusts how much influence the LoRA layers have on the model's final predictions. (default: 16).

### 8.6.7 LoRa target modules

By default, **LoRA target modules** are set to null.

### 8.6.8 LoRa Task Type

The task type for LoRa adaptation is CAUSAL_LM by default.




## 8.7 Model Saver Section

### 8.7.1 Max number of saved models on epoch end

Specifies the maximum number of models to be saved during epoch cycles. If the number of saved models exceeds this value, the earliest saved model will be removed. Set to -1 to retain every model saved per epoch. (default: -1)

### 8.7.2 Enable save model on iteration

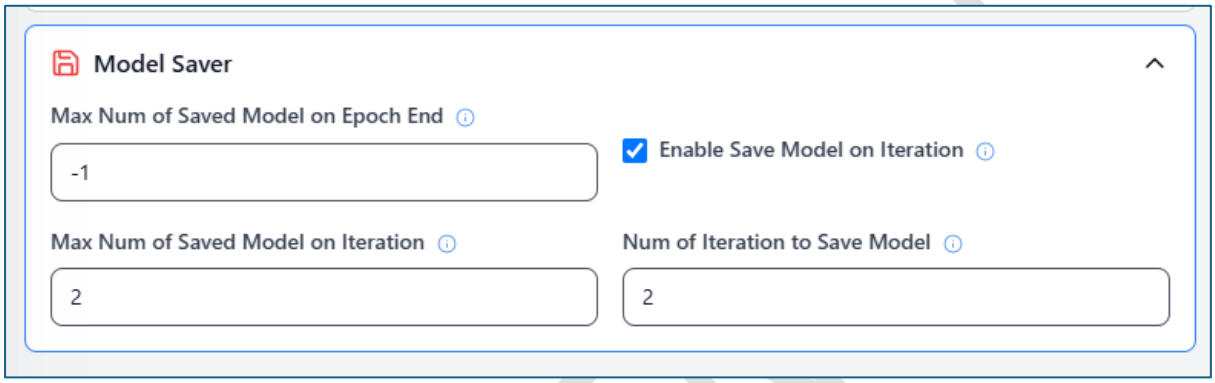
Specifies whether to save the model during iteration cycles. (default: false)

### 8.7.3 Max number of saved models on iteration

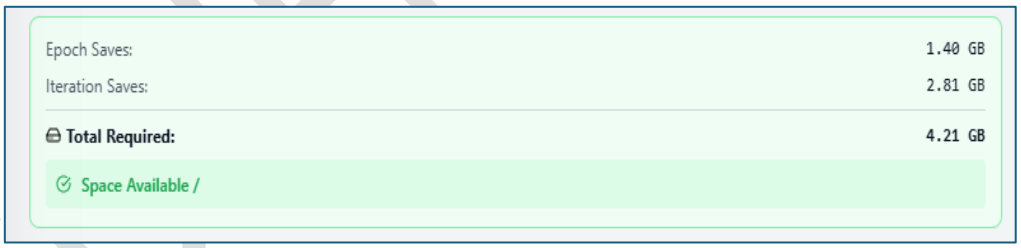
Specifies the maximum number of models to be saved during iteration cycles. Must be greater than 0. It is recommended to set at least 2 to avoid potential loss due to device issues. (default: 2)

### 8.7.4 Number of iterations to save model

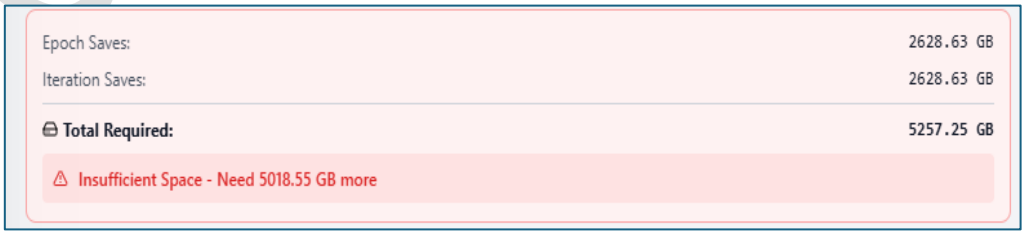
Specifies how often a model should be saved per number of steps. Must be greater than 0 and a multiple of gradient_accumulation_steps. (default: 2)




- Automatically calculates the required storage based on the selected model, output directory, epoch saves, and iteration saves.
- Displays total estimated space needed in GB.
- Checks available disk space in the output directory.

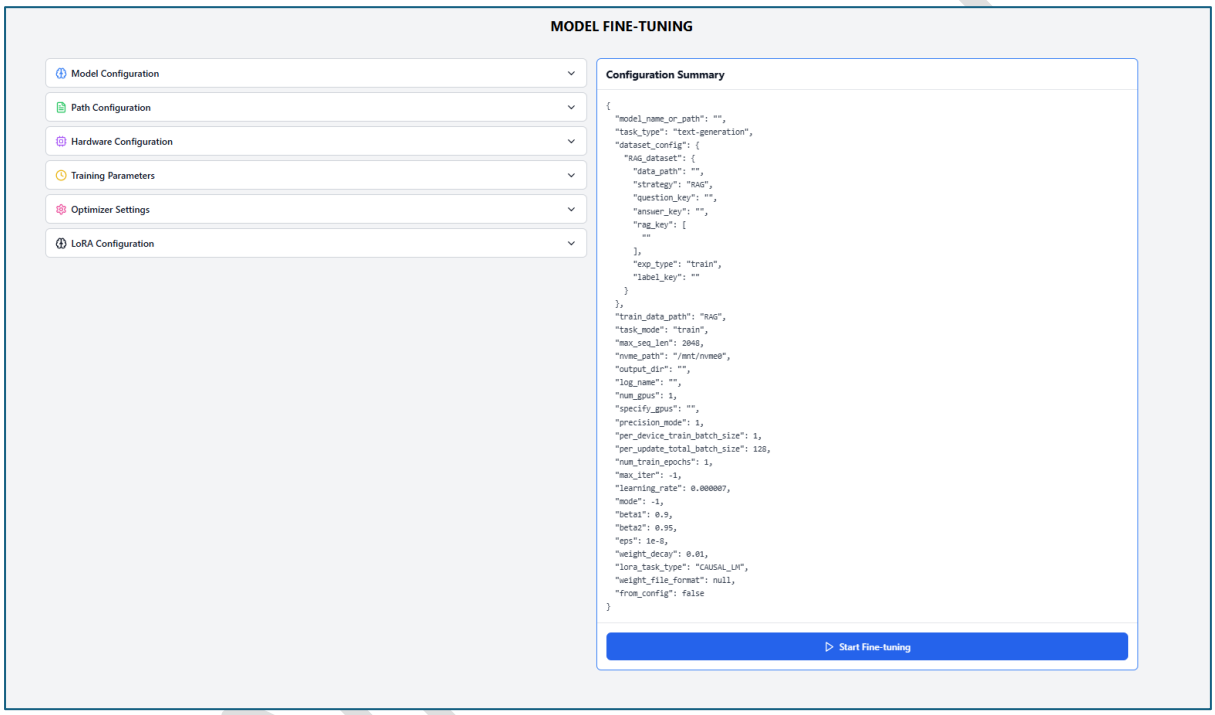



- If insufficient space is detected, **Start Finetuning** will remain disabled.



## 8.8 Configuration Summary

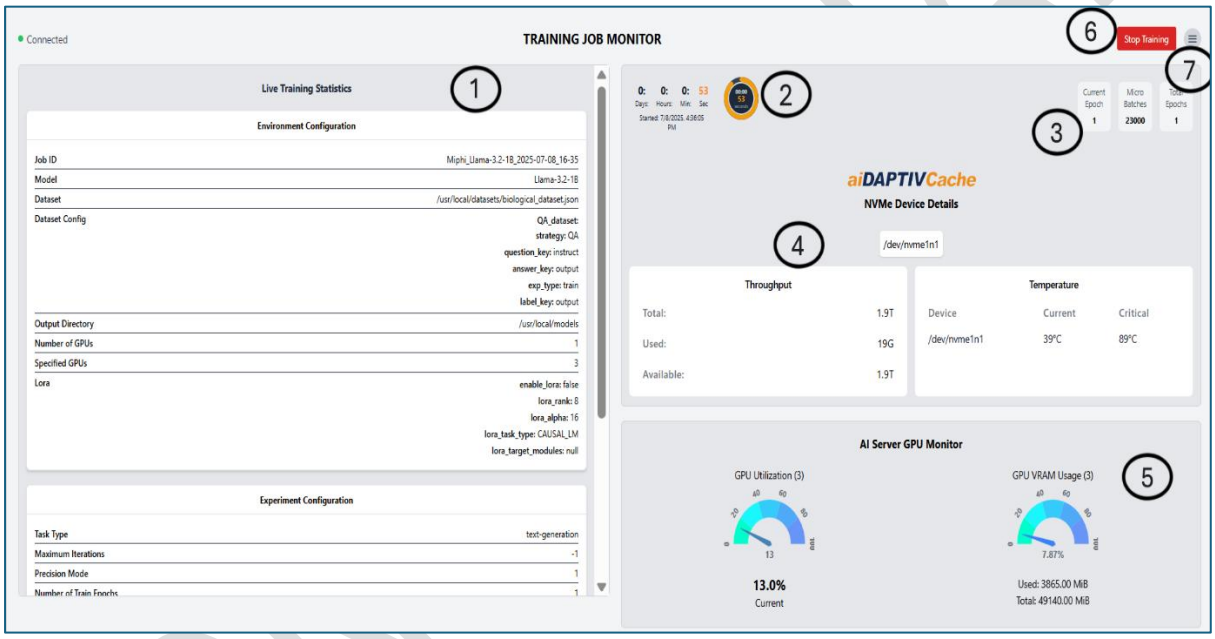
- After setting all parameters, they will automatically update in the **Configuration Summary**, where you can review all the values you have provided.
- Once everything is confirmed, click the **Start Fine-Tuning** button to begin the process.
- The system will then redirect you to the **Monitor** page, where you can track the training progress in real time




**Note:** Before starting fine-tuning, ensure that all parameters are correctly updated to avoid errors

# 9. MONITOR

The **MiPhi Monitor** offers a real-time overview of the training process, providing key insights into both system performance and training metrics. It displays training configurations, including NVMe cache details, device usage, elapsed time, current epoch, total epochs, and microbatches. The page tracks GPU utilization, VRAM, CPU, and RAM usage, ensuring efficient resource monitoring. Users can access and download training logs for analysis, view a loss vs. timestamp graph, and examine detailed GPU statistics. The training process can be stopped at any time, and the sidebar allows quick access to the latest five training jobs for efficient session tracking.




## 9.1 Training Statistics

The **Training Statistics** section provides configuration details for training jobs and dynamically updates based on the training status. It is divided into two parts:

- **Environment Configuration**: Displays metadata such as Job ID, model name, dataset, dataset configurations , output directory, GPU settings, and LoRA parameters.
- **Experiment Configuration**: Shows training-specific parameters including task type, number of epochs, batch sizes, sequence length, triton, optimizer settings, learning rate scheduler, and precision mode.

Behaviour:

- If **no training** has been run yet, the section shows **"No process running."**
- If a training job is **in progress**, this section displays **live statistics** of the current job.
- Once a job is **completed**, the section shows details of the **latest completed job** along with its Job ID.

It helps to easily track and reference the most recent training configurations and progress.

## 9.2 Elapsed Time

The **Elapsed Time** section provides a real-time display of the duration since the current training job began. Time is presented in a structured format—**days, hours, minutes, and seconds**—accompanied by a visual circular progress indicator for enhanced clarity. Additionally, the exact start timestamp of the training session is shown below the timer for precise reference.

This section becomes visible **only when a training process is actively running**, allowing to monitor the session duration as it progresses. When no training is in progress, the section remains hidden.

## 9.3 Epoch and Micro-batch

This section displays real-time training metrics, including:

- **Current Epoch**: The epoch currently in progress.
- **Total Epochs**: The total number of training epochs defined.
- **Microbatches**: The cumulative number of microbatches processed within the current epoch.

This panel is visible **only while a training session is active**, providing clear visibility into training progress at both the epoch and batch level.

## 9.4 NVMe Device Details

This section provides detailed monitoring of the NVMe storage device actively used during training. It is designed to offer critical insights into both the storage capacity and thermal performance of the device in real time.

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- **Device Path**: Displays the device path used of the active NVMe mount point (e.g., /dev/nvme2n1).

**Throughput**

- **Total** – The total capacity of the NVMe device.
- **Used** – The amount of storage currently occupied by training processes or other operations.
- **Available** – Remaining free space available on the device.

**Temperature**

- **Current** – Real-time operating temperature of the NVMe device.
- **Critical** – The maximum temperature threshold before the device may throttle or become unstable.

## 9.5 GPU Monitoring

The **GPU Monitoring** section displays real-time metrics of the GPU actively used for training. It includes:

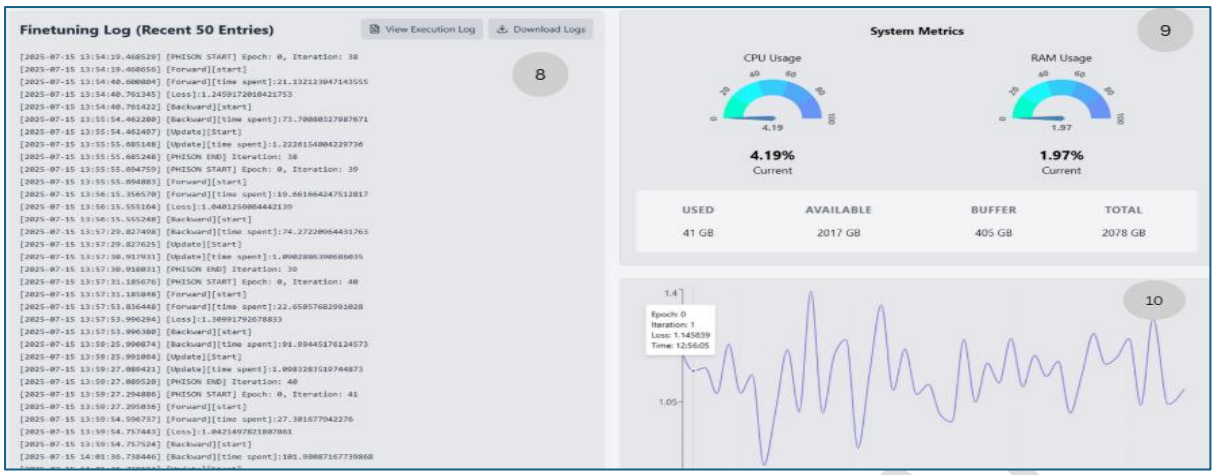
- **GPU Utilization**: Shows the current processing load (%) on the training GPU.
- **GPU VRAM Usage**: Displays the used and total memory (MiB), along with the percentage of VRAM consumption.

## 9.6 Stop Training

This section displays a **Stop Training** button when a training job is in progress. Clicking this button will immediately terminate the active training process.

- When **training is active**, the button is visible and functional.
- When **no training is running**, the section displays **"No process running."**

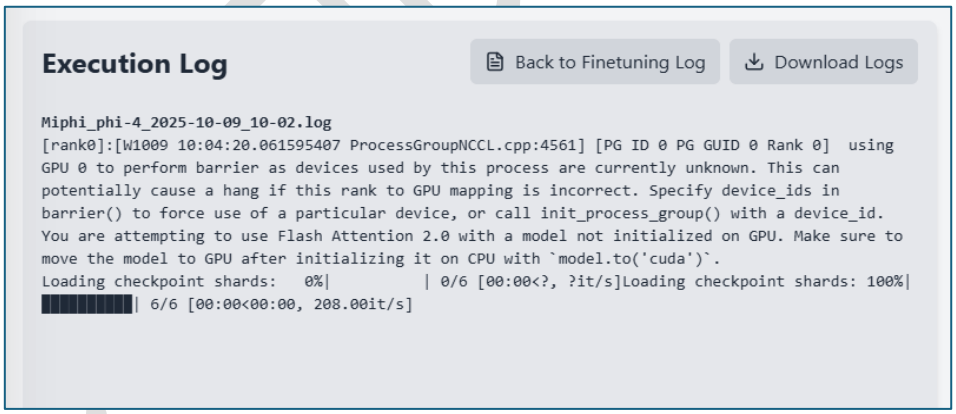
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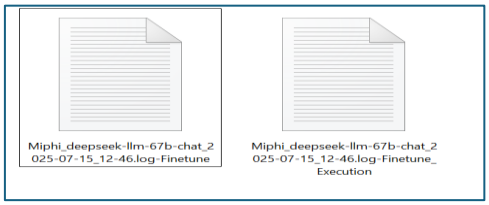

## 9.7 Training Logs

This section displays the **latest 50 log entries** from the current training session in real time.

- When training is **active**, logs update live to show forward passes, backward passes, loss values, iteration details, and time spent.
- When training is **inactive**, it displays logs from the most recent completed session.
- The **View Execution Log** button allows users to view the **execution logs** specific to that particular training job for detailed operational insights.




- The **Download Logs** button allows users to download both the **fine-tuning log and execution log** for the selected training session. The logs are downloaded together for further analysis or archival.




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## 9.8 System Metrics

This section provides real-time insights into system resource usage:

- **CPU Usage**: Displays the current percentage of CPU utilization.
- **RAM Usage**: Shows the current memory consumption as a percentage.
- **Disk Summary**: Indicates used, available, buffer, and total disk capacity.

## 9.9 Training Graph

This section visualizes the **loss over time**.

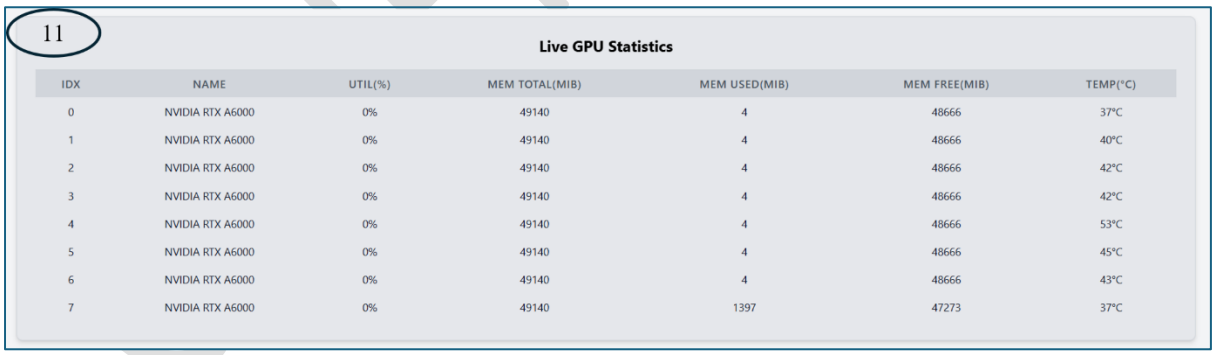
- **When training is active**, the graph updates in real-time, reflecting live loss values.
- **When training is inactive**, it displays the most recent complete loss graph from the last fine-tuning session.

## 9.10 Live GPU Statistics

This section provides **real-time monitoring** of all available GPUs, displaying:

- **Utilization (%)**
- **Total, used, and free memory (MiB)**
- **Temperature (°C)**

If a GPU's temperature exceeds **93°C**, it is highlighted with a **red background** to indicate potential overheating.



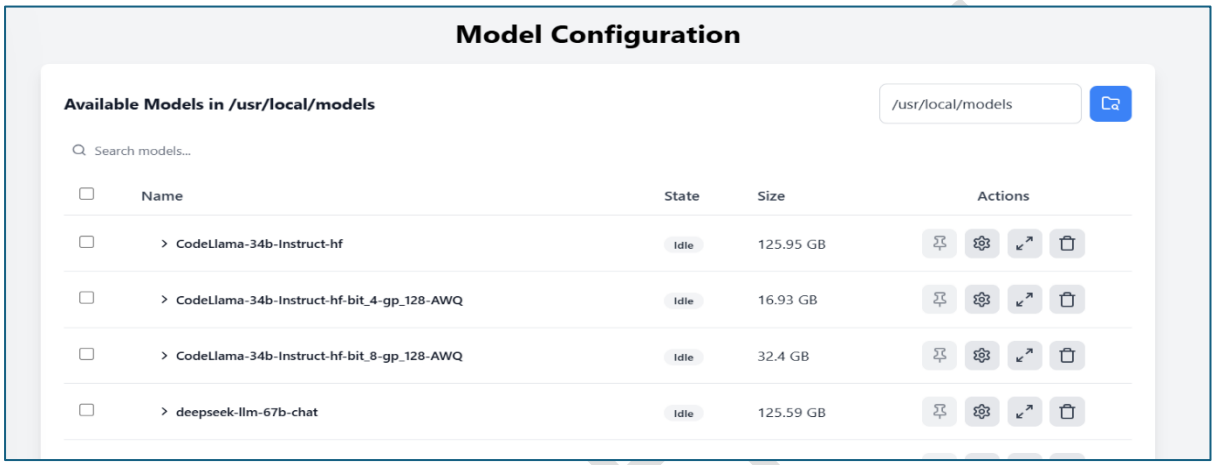

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# 10. INFERENCE

## 10.1 Model Configuration

### 10.1.1 Select Model for Inference

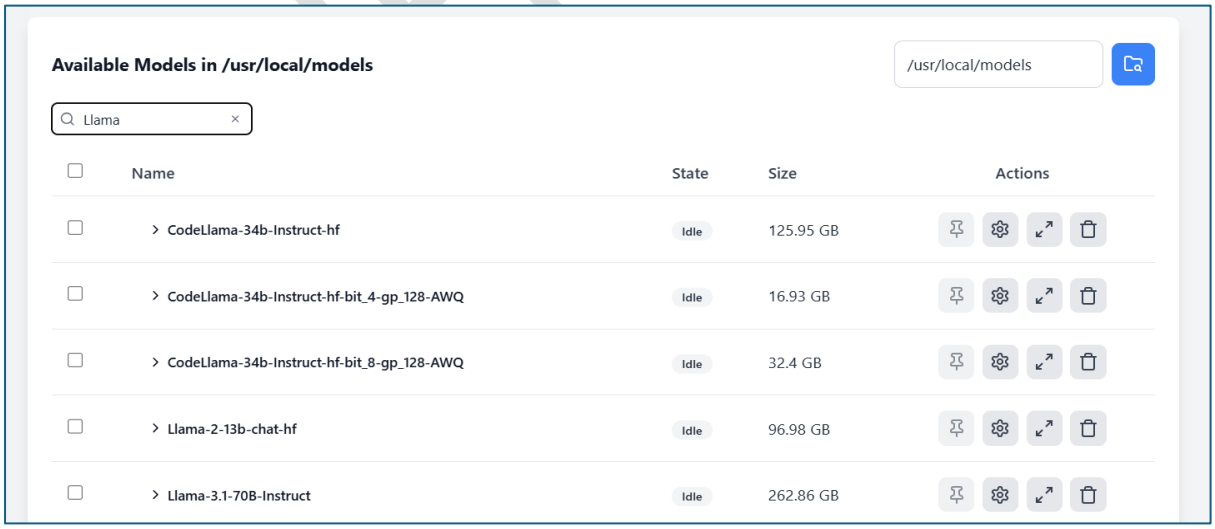
- This page displays all available models based on the specified model directory path.
- For optimal performance, it is recommended to load the model using a number of GPUs that is a power of 2 (e.g., 1, 2, 4, 8).




- Here you can give the custom directory path where the models are located.




- Use the search bar to quickly filter and find models by typing their name



### Components of Model Table

| Column | Description |
|---|---|
| Name | The name of the model (e.g., Llama-3.1-8B-Instruct, Gemma-2-9B-It) |
| State | Status of the model (Idle or Running) |
| Size | Size of the model in gigabytes (GB) |
| Actions | Buttons for model operations |

### Model Actions

For each model, the following actions are available (represented by icons in the Actions column):

- **Pin Model** : Load and pin a model for use
- **Settings** : Configure model parameters
- **Expand** : View and perform model quantization
- **Delete** : Remove the model from the system

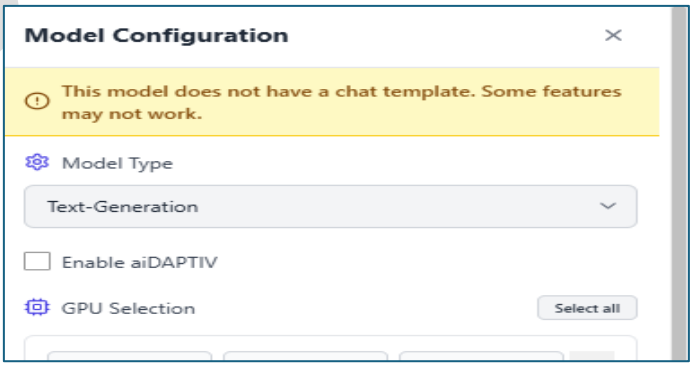
### 10.1.2 Model Configuration

- Click the settings button to adjust model parameter inference.

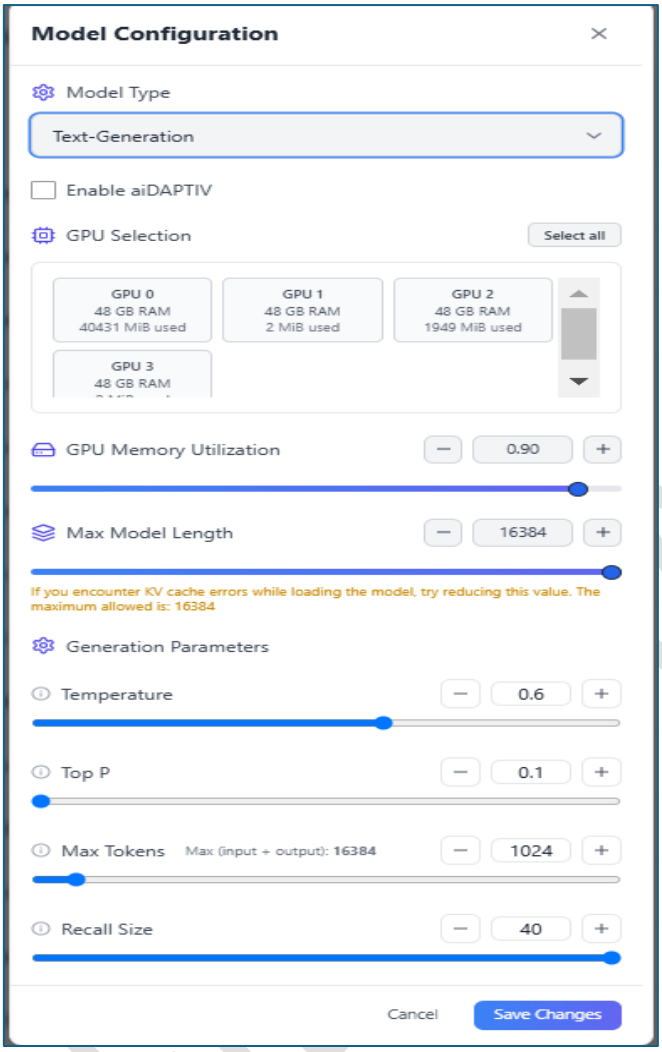



- Now the model configuration box will appear. Select the required values to load model and click ‘Save Changes’.

  **Important:** You must have a chat template of the model configured. Otherwise, you will not be able to save changes or load a model.




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**Adjusting Model Parameters**

You can modify the model's behavior by adjusting the following parameters:

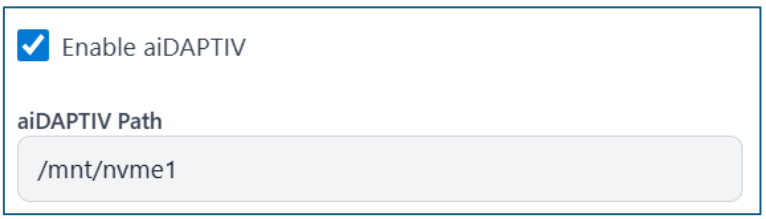
**Supported Model Type**

- Text-Generation
- MultiModel Text-Generation
- Zero-shot Image classification

**Enable aiDAPTIV ( Supported only for Text-Generation )**

- Uses the aiDAPTIV cache to load the model for inference.

**Note:** When the model is unloaded, the corresponding aiDAPTIV cache data stored in the specified path will be **automatically deleted** to free up space.



**GPU Selection**

- Select the specific GPUs for loading the model.
- Displays GPU index, total RAM, and currently used memory.

**GPU Memory Utilization**

- Sets the percentage of GPU VRAM that can be utilized for loading the model.
- **Default:** 0.90
- Use the slider or +/- buttons to adjust.

**Max Model Length**

- Sets the maximum model length allowed for loading.
- Range: 1 to [Model's maximum context length]
- The maximum allowed length is determined by the model's architecture and configuration.
- For most models, it is up to 4096 tokens, but check your specific model's documentation or config.
- If you encounter KV cache errors while loading the model, try reducing this value.

**Generation Parameters**

- **Temperature:** Controls randomness/creativity of responses (default: 0.6)
  - Range: 0.0 to 1.0
  - Lower values (closer to 0) make responses more deterministic and focused
  - Higher values (closer to 1) make responses more creative and diverse
- **Top P:** Controls nucleus sampling threshold (default: 0.1)
  - Range: 0.1 to 1.0
  - Works alongside temperature to influence response variety
- **Max tokens:** Controls the maximum length of generated responses
  - Default: 1024
  - Range: 1 to (Max Model Length - Prompt Length)
  - The effective maximum is subject to the model's maximum context length and current input prompt size.
  - Use slider or +/- buttons to adjust.
  - Higher values produce longer responses.

⚠️ **Warning:** To generate a response, the combined input and max tokens should stay under the allowed limit. If exceeded, shorten your max tokens.

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- **Recall Size:** The value for reading chunks in file (default: 40) while using RAG.
    - Use slider or +/- buttons to adjust

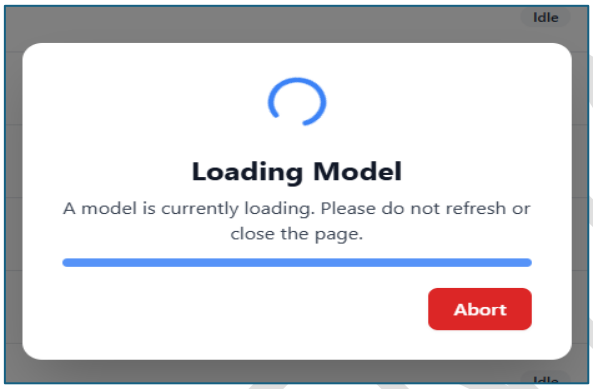
Finally, configure the model parameters and save the changes.

### 9.1.3 Pin the Model

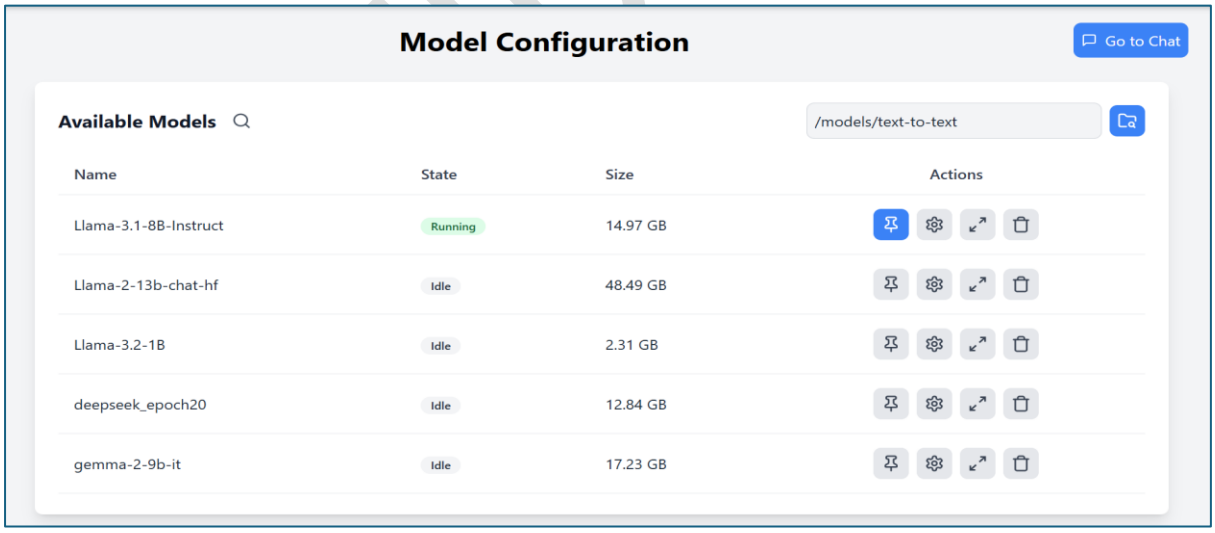
- Click the pin button to load the model and prepare it for inference.




- The model may take a few moments to load.




- You can abort the loading process by clicking on '**Abort** button**.**
- The selected model will be pinned and appear on the top of the model table. The model is ready for inference.

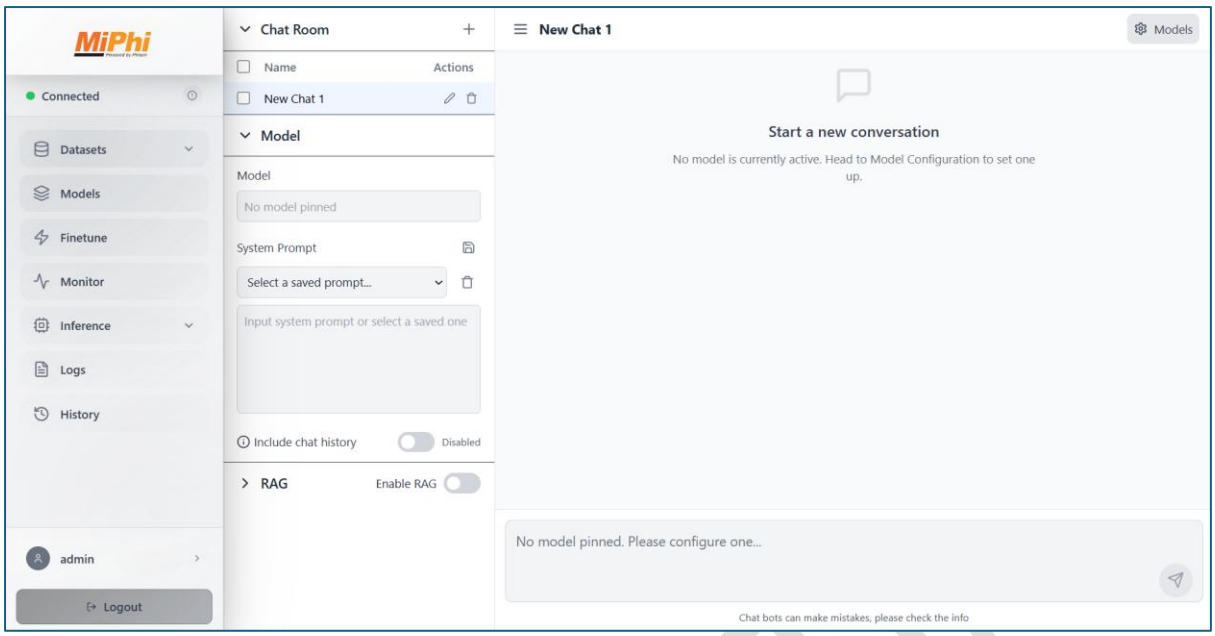



## 10.2 Chat Room

### 10.2.1 Navigate to Chat Bot

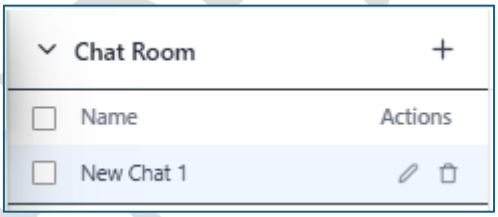
- Click 'Go to Chat' or visit the Chat-Room page to start interacting with the model.

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### 10.2.2 Create New Chat

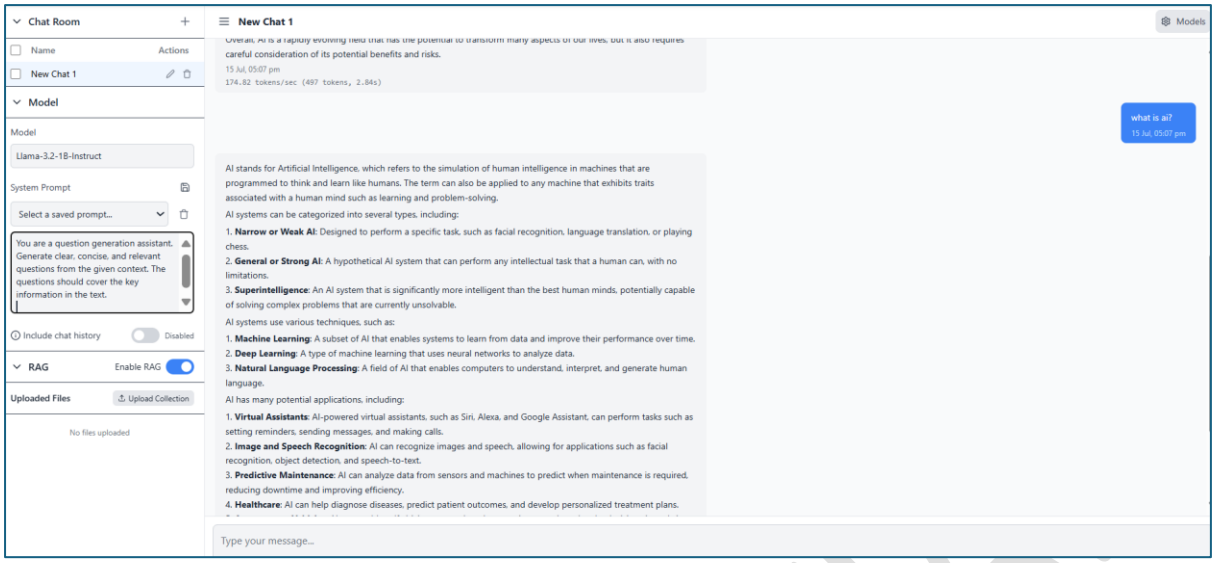
- You can create separate chat sessions for multiple queries based on requirement for the loaded model by clicking the '+' icon in the Chat Room.
    - You can also rename the chat session title or delete the chat session as needed.




### 10.2.3 System Prompt

The **system prompt** is a predefined instruction given to the model before user input. It establishes the behaviour, tone, and role of the model throughout the interaction. This prompt ensures consistent responses and helps guide the model's output according to the intended use case.

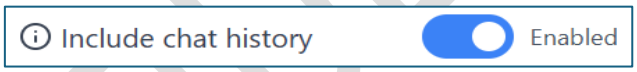
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**Note:** You can save system prompts for future use. You can select a saved prompt from the dropdown, use it, or delete it if no longer needed.

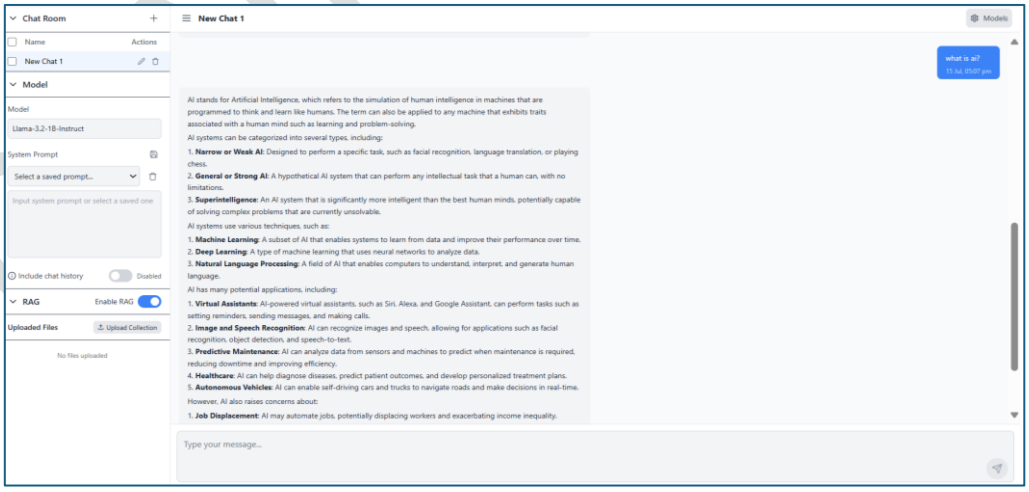
### 10.2.4 Include Chat History

- When enabled, the system includes previous chat messages to provide better and more context-aware responses.
- If you do not want to use chat history, you can simply disable this option.

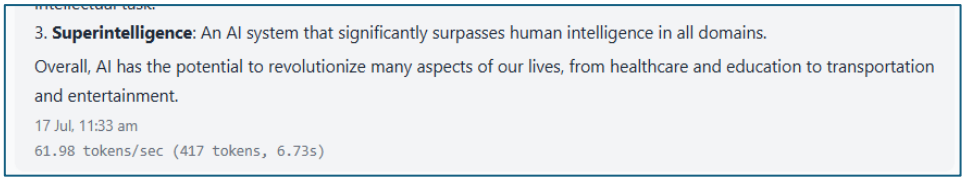



### 10.2.5 Start Your Conversation

All set! You can now start interacting with the model by asking questions or sharing ideas.



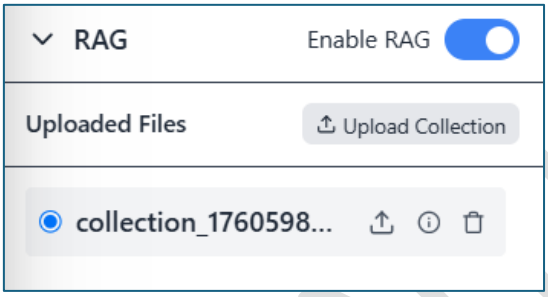
- Displays the **generation speed**, showing how many tokens were generated per second. Useful for monitoring model performance and efficiency.




### 10.2.6 RAG (Retrieval-Augmented Generation)

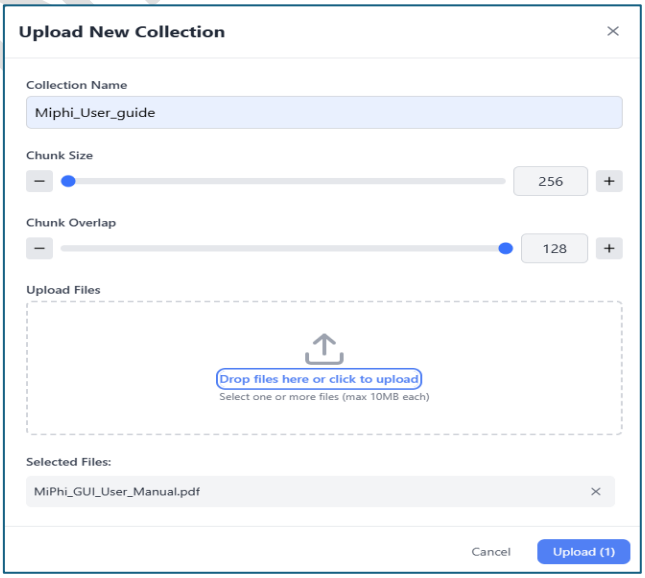
RAG helps the model give better answers by first finding relevant information from documents and then using it to generate a response.

- **Enable the RAG toggle** to activate retrieval-augmented generation.




**Upload files to create a Collection**

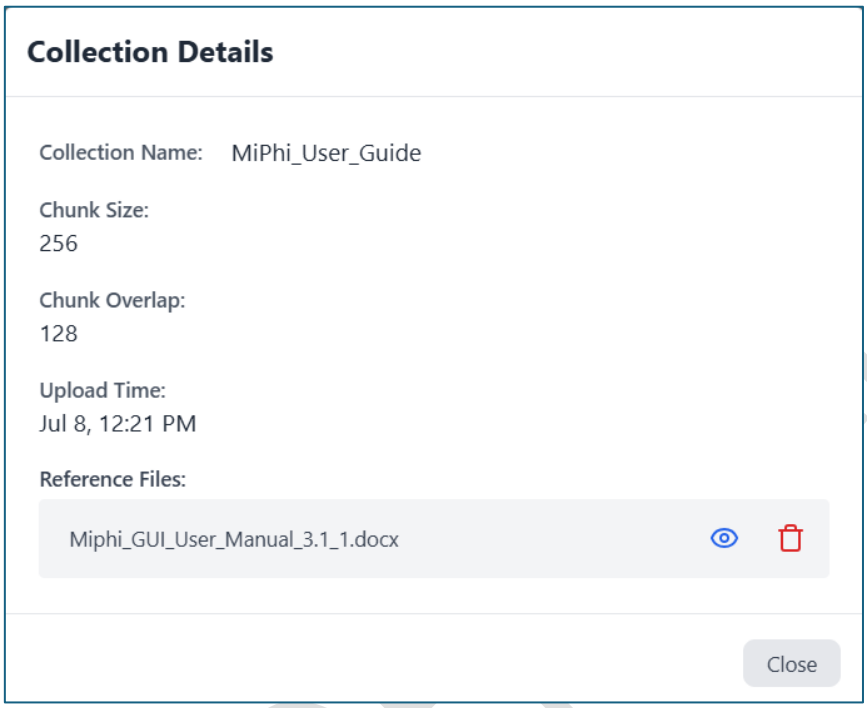
- **Collection Name**: Enter a custom name for your collection. If left blank, a default collection ID will be assigned automatically.
- **Chunk Size:** The amount of data contained in each chunk when processing retrieved documents. Range: [256, 2048]
- **Chunk Overlap:** The amount of data that overlaps between consecutive chunks when processing. Range: [0, Chunk Size * 0.5]
- **Upload File Area**: File upload area for output Collection (can upload multiple files)




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Function description:

- Click the **Info** button to view details about the collection, including its name, number of files, chunk size, overlap, Upload Time.




- : Click the **Delete** icon to permanently delete the collection and all associated files.
- : Click the preview icon to preview the particular file.

## Multimodal Text-Generation

This section explains how to configure and load **Image-Text-to-Text** and **Automatic Speech Recognition (ASR)** models using the **Multimodal Text-Generation** option.

### Select the Correct Model Type

To work with **images, audio, and text together**, you must use a multimodal model.

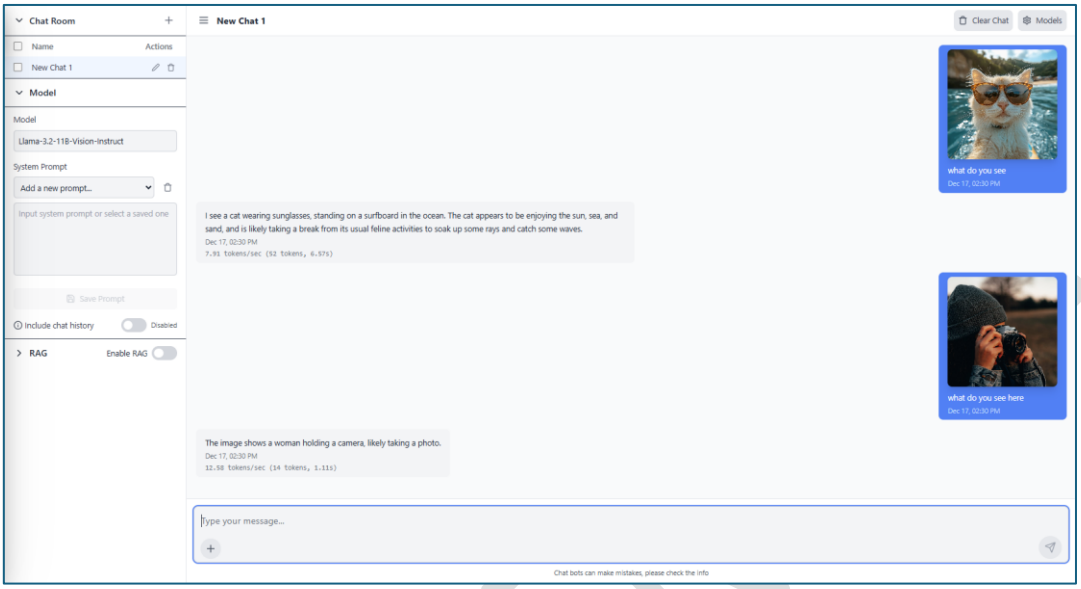
**Steps:**

1. Select **Multimodal Text-Generation** in **Model Type**, choose the required GPUs, configure **GPU Memory Utilization**, **Max Model Length**, and **Generation Parameters** , then load the model.
2. Open the **Chat Room** and upload an **image** for the image-text-to-text model and provide the related text input along with the image; if using an **ASR model**, upload an **audio file** and specify the **language** you want to transcribe.
3. Click **Send** to receive the multimodal output (image-text-to-text or speech-to-text).

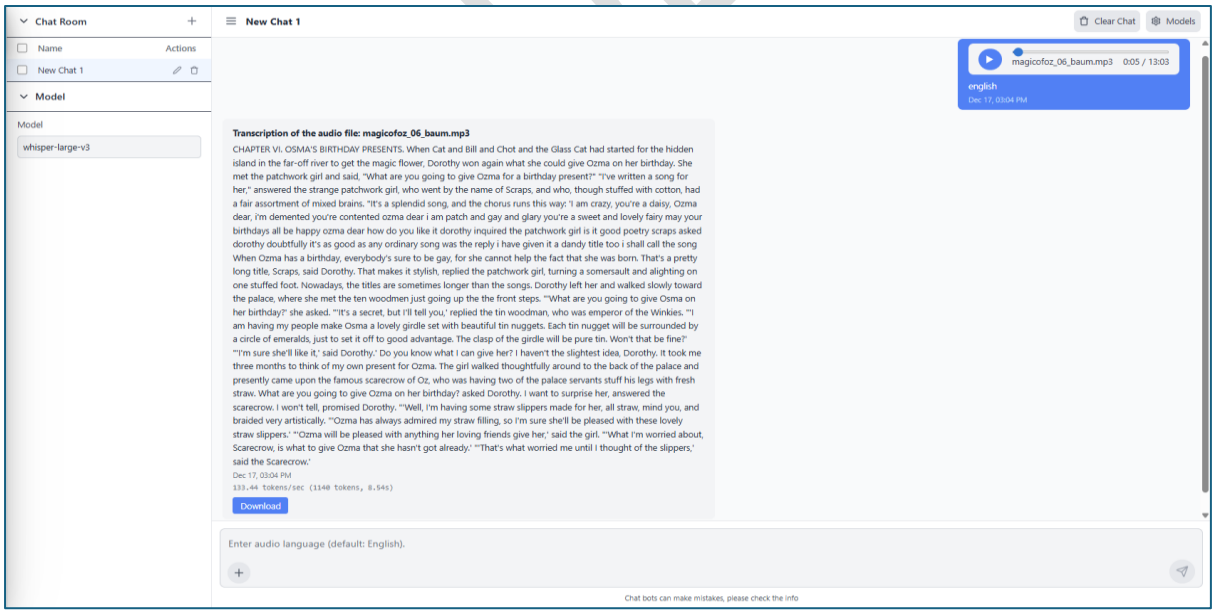
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**Note:** If you encounter a *Max Model Length* error while loading the model, reduce the max model length value and try loading the model again.

**Image-Text-Text Interaction Example:**

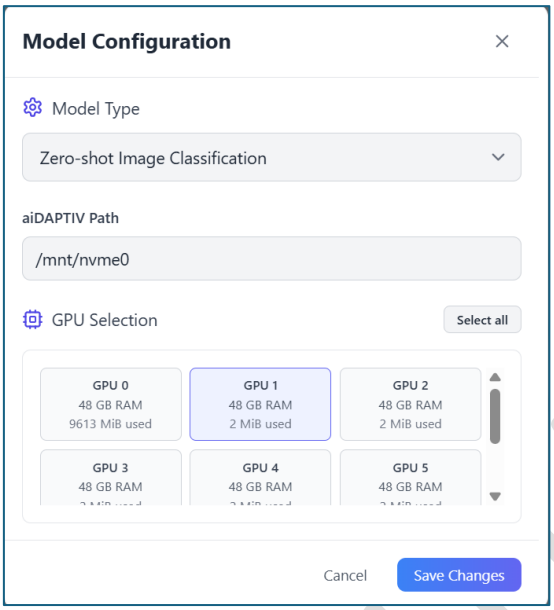



**Automatic Speech Recognition (ASR) Interaction Example**

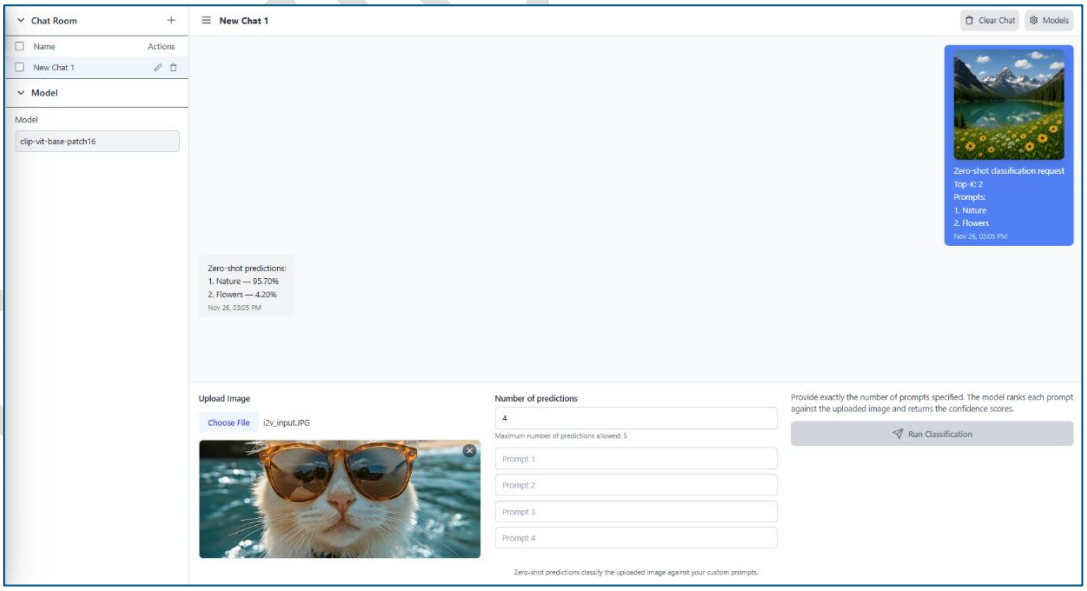



Upload an audio file and enter the language you want the speech to be transcribed into.

## Zero-shot Image Classification




- Select **Zero-shot Image Classification** in Model Type, choose GPUs, and load the model.
- Go to the **Chat Room** and upload any supported image.
- Enter the number of predictions (Top-K) and provide the same number of prompts. (maximum number of predictions allowed:5)
- Click **Run Classification** to get ranked zero-shot results with confidence scores



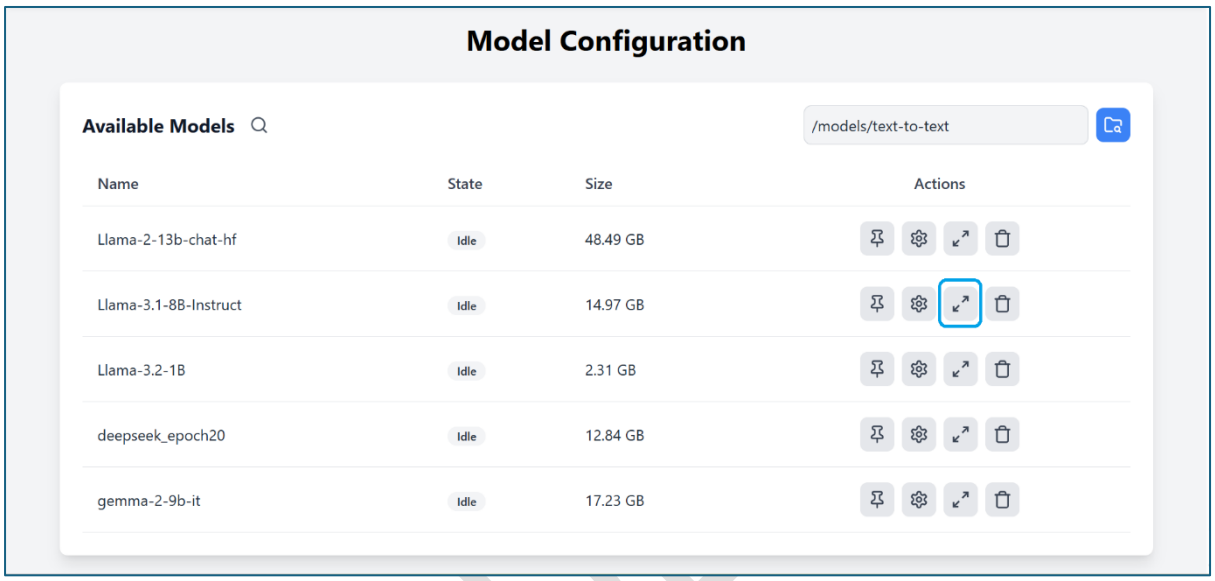

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## 10.3 Model Quantization

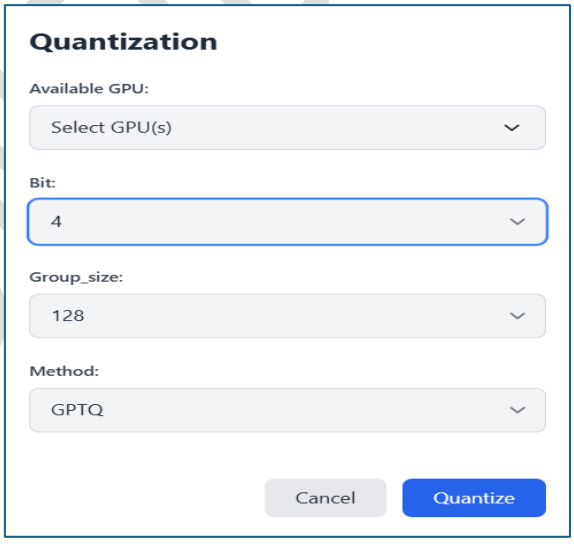
Model quantization is a technique used to reduce the precision of the numbers used to represent a model's parameters, such as weights and activations, from higher precision formats like FP32 (32-bit floating point) to lower precision formats like INT8 (8-bit integer) or even INT4.

### 10.3.1 Model Parameters for Quantization

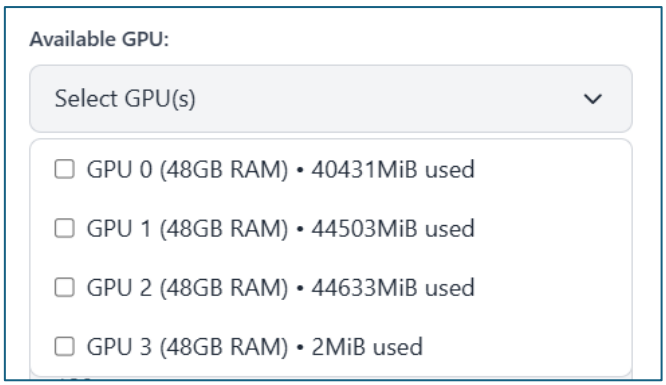
- Click the settings button to adjust model parameters.




- Now the quantization option box will appear.




- **Available GPU:** Select specific GPUs from the available list of GPUs you want to use for quantization from the drop-box. You can select one or multiple GPUs.



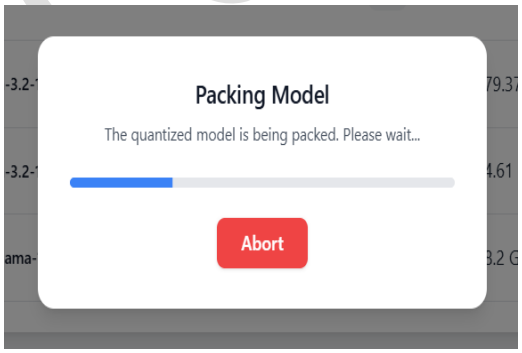

- **Bit:** The bit value to quantize the model in lower precision data. (default: 4)
- **Group Size:** Specify the group size used during quantization.
- **Method:** Choose the quantization method. Options include **GPTQ** and **AWQ**.

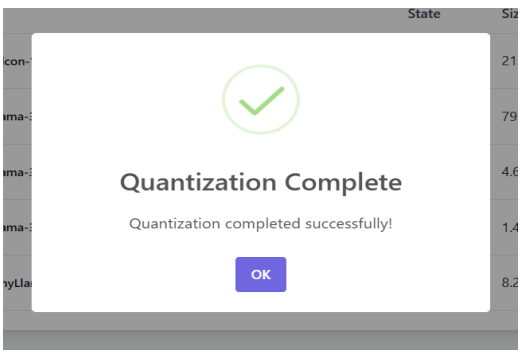
After selecting all the options click “Yes” to start model quantization.

### 9.3.2 Monitor Quantization Progress

- You will get the below window when the model quantization starts. In the bottom a black progress bar appears**.**



The progress bar has the following components:

- Quantization Progress: Left side is the real time progress percentage of model quantization
- Quantized model path: Central is the quantizing message along with the name of quantized model
- Cancellation: Right side is having one “Cancel” button to stop quantization. It will show one warning message after clicking “Cancel” to confirm whether you want to stop quantization.

After clicking “OK”, it will cancel quantization and confirmation message will appear.

- When the quantization is completed successfully, the quantized model will appear in the model configuration page of the same directory. The quantized model name is followed by the group size and bit value.



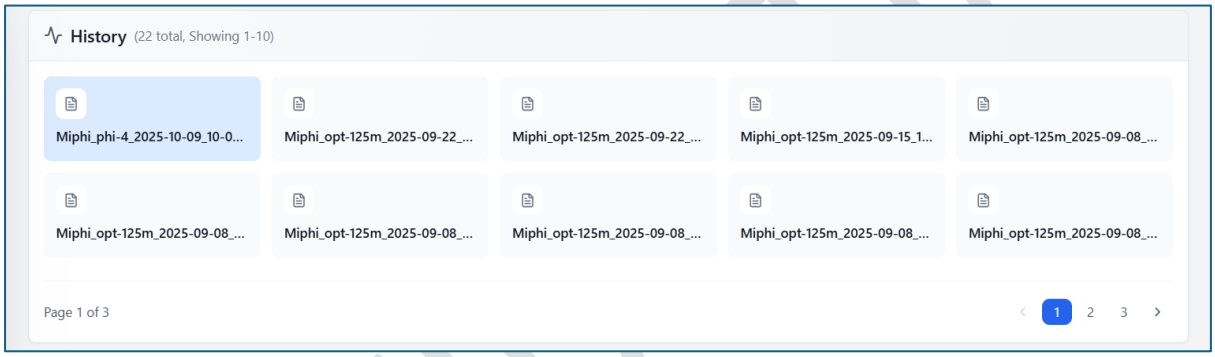

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# 11. HISTORY

The **MiPhi History** Page displays a complete list of all previously fine-tuned jobs. It allows users to select any past fine-tuning session and view its detailed information, including configurations, storage usage, GPU utilization, logs, and performance graphs.

## 11.1 Jobs List

- The top section contains a **list of all completed fine-tuning jobs**.
- Each job entry is represented with a **log filename** (for example: Miphi_phi-4_2025-10-09_10-02.log).
- Users can **click on a log file** to load and view its full details.
- Pagination at the bottom helps navigate through multiple pages of logs.

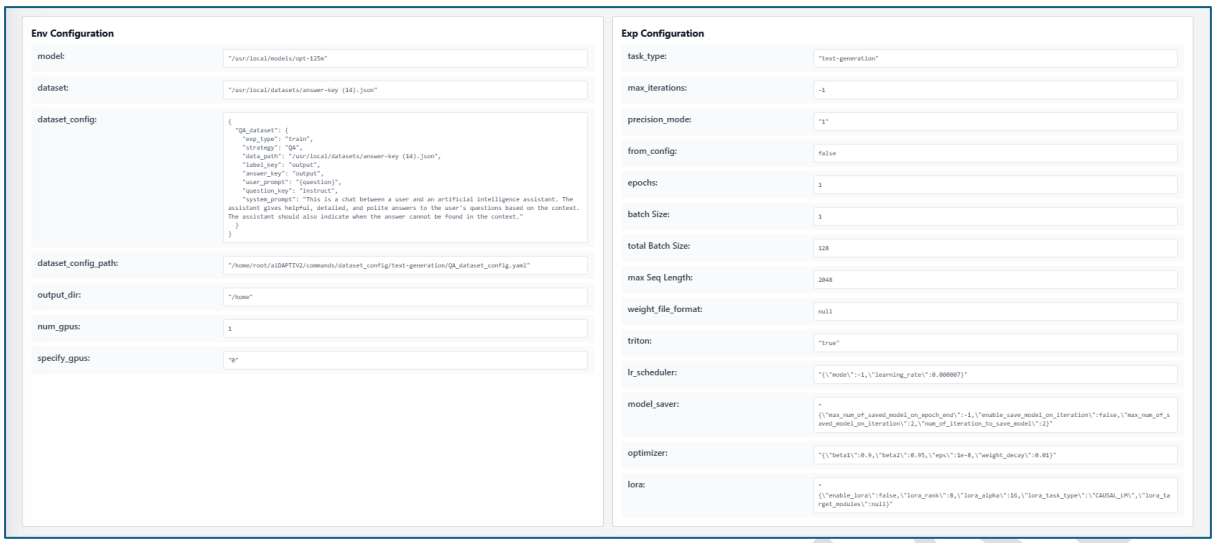



## 11.2 Finetuning Configurations

Once a job is selected, its configurations are shown in detail:

- **Environment Configuration**: Includes model path, dataset, dataset config path, output directory, and GPU setup.
- **Experiment Configuration**: Displays fine-tuning parameters such as task type, number of iterations, precision mode, batch size, sequence length, scheduler, optimizer, and loss type.
- These configurations help reproduce or analyze the exact experimental setup used for fine-tuning.

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## 11.3 Resource Utilization

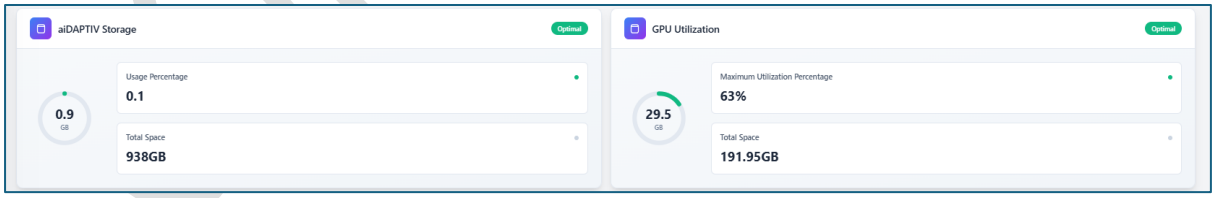
Displays key resource statistics for the selected fine-tuning session.

**aiDAPTIV Storage** shows how much storage space was used, including **usage percentage** and **total available capacity** (e.g., *0.1% of 398 GB*).

- Helps monitor **disk utilization** and manage **saved model checkpoints** efficiently.

**GPU Utilization** provides details on **maximum GPU usage (selected GPU)** and **total system GPU memory available** (e.g., *63% of 191.59 GB*).

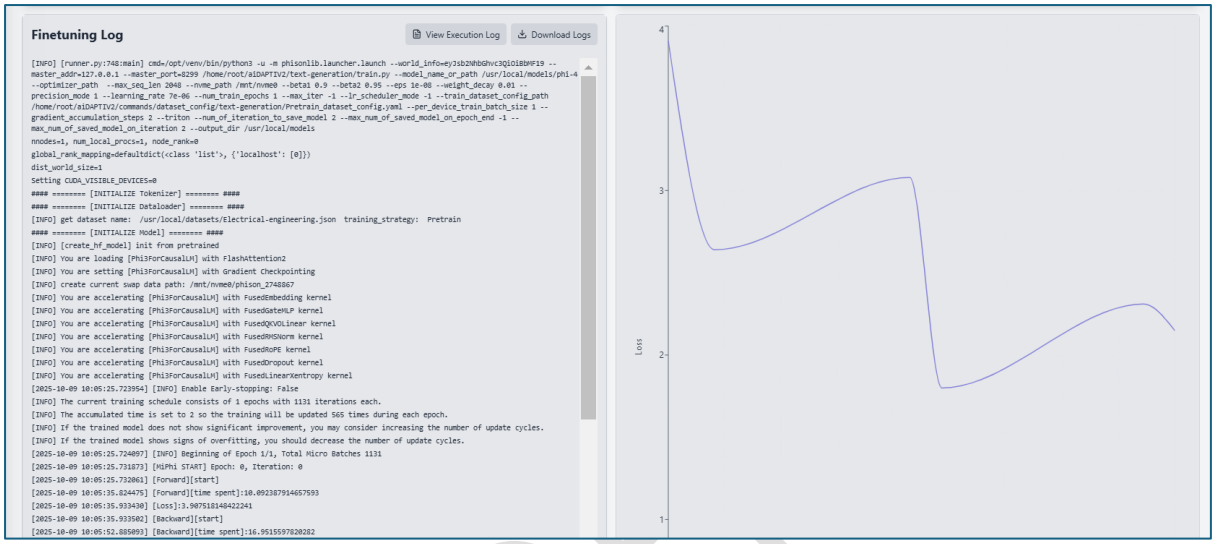
- Useful for analyzing **hardware performance**, understanding **resource consumption**, and optimizing **future fine-tuning runs**.




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## 11.4 Training Log and Graph

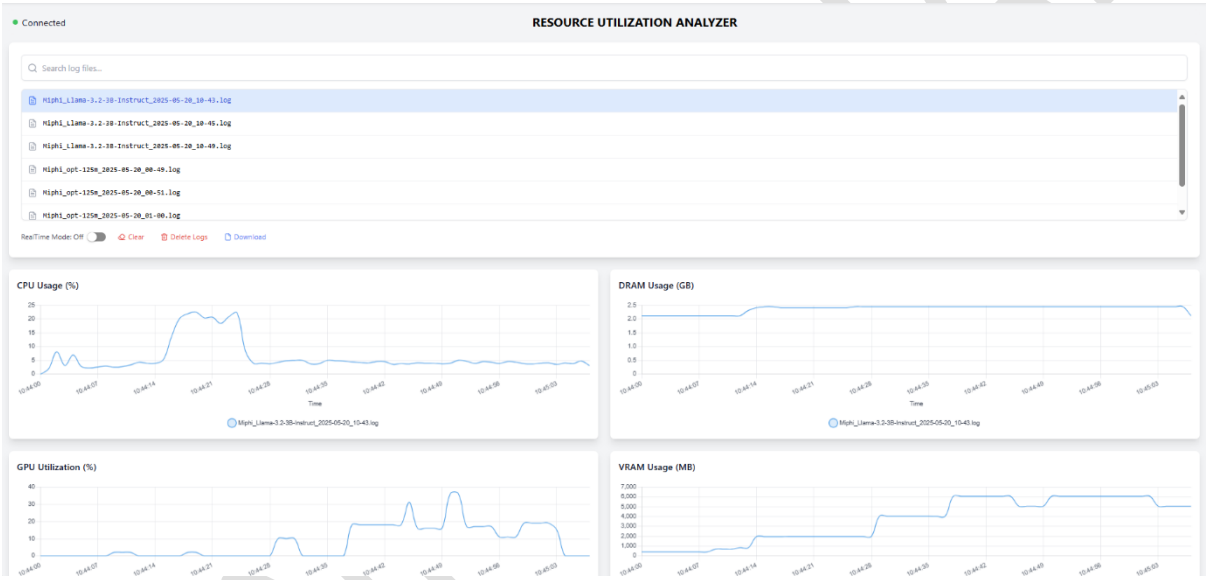
- Displays the **training logs** and the **corresponding performance graph** for the selected fine-tuning session.
- The **graph panel** visualizes the training progress over time, helping users understand how the model performed and stabilized during fine-tuning.

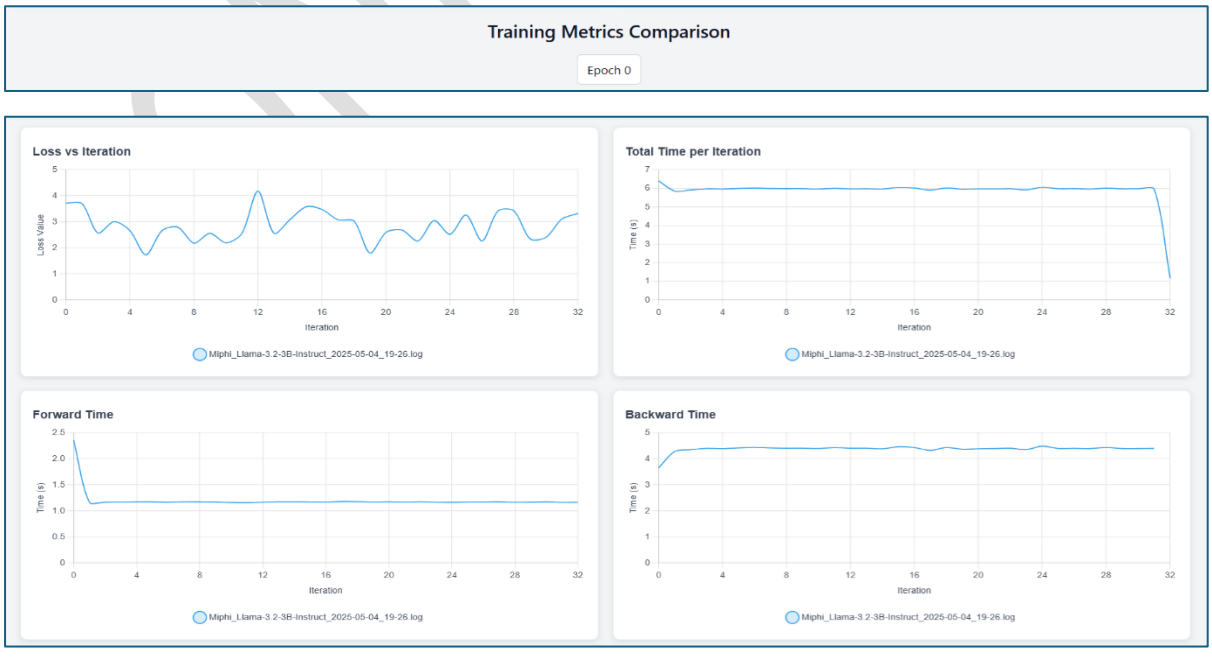


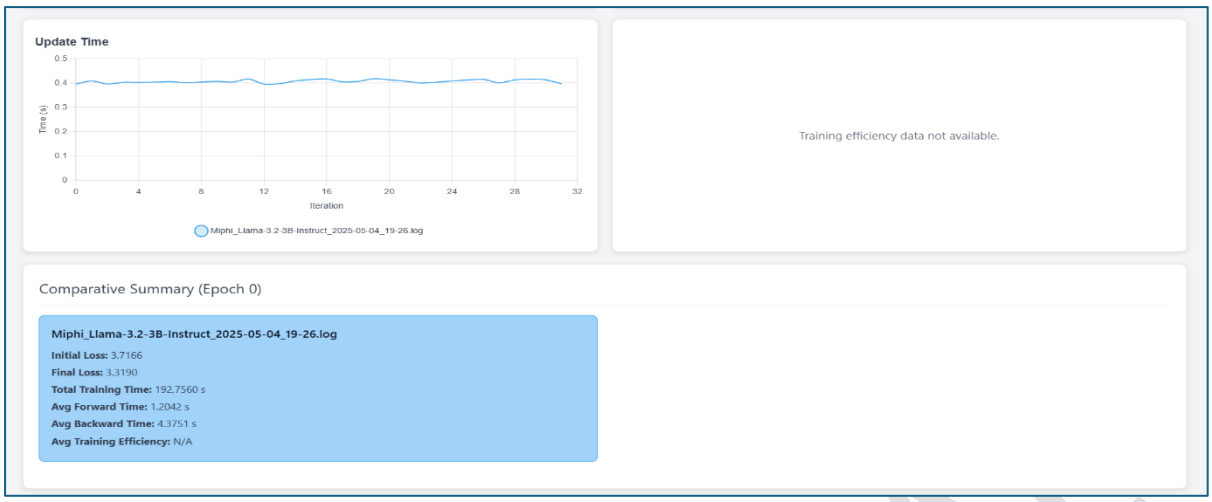

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# 12. LOGS

**MiPhi Logs** provides a centralized dashboard for monitoring and comparing training performance across multiple fine-tuning jobs. Users can select up to six log files to visualize key metrics, including CPU and RAM usage, GPU utilization and VRAM, NVMe usage and temperature, loss vs. iteration, and iteration time breakdown (forward, backward, update). The platform offers a comparative summary across all epochs, a real-time analysis toggle for active training sessions, and options to download or clear selected logs. Users can also search through logs and delete individual or multiple files, enabling efficient tracking, diagnosis, and performance comparison of fine-tuning processes.






## 12.1 Log File Selection

This section allows users to select and compare up to six log files simultaneously. Upon selection, the dashboard displays system metrics (CPU usage, RAM usage, GPU utilization, VRAM usage, NVMe usage, and temperature) and training metrics (loss, forward time, backward time, update time, total iteration time, and training efficiency).

- For system metrics, the x-axis represents **time** when viewing a single epoch and **iterations** when comparing across multiple files.
- For training metrics, the x-axis is always **iterations**.

Users can also filter metrics by epoch for focused analysis.

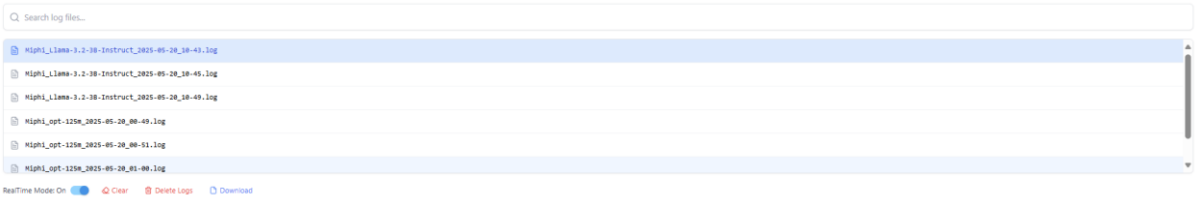
**Additional Features:**

- Users can filter metrics by **epoch** for focused analysis.
- A **searchable dropdown** allows users to quickly find log files.
- By default, the dropdown shows only the **latest 10 log files**, streamlining the selection process and improving performance.

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## 12.2 Real-Time Analysis

The **Real-Time Analysis** toggle allows users to monitor live system and training metrics during an active fine-tuning process. When enabled, it dynamically fetches and updates graphs in real time, providing up-to-date insights into performance, resource usage, and training behavior. This feature is only active when the selected training process is currently running.

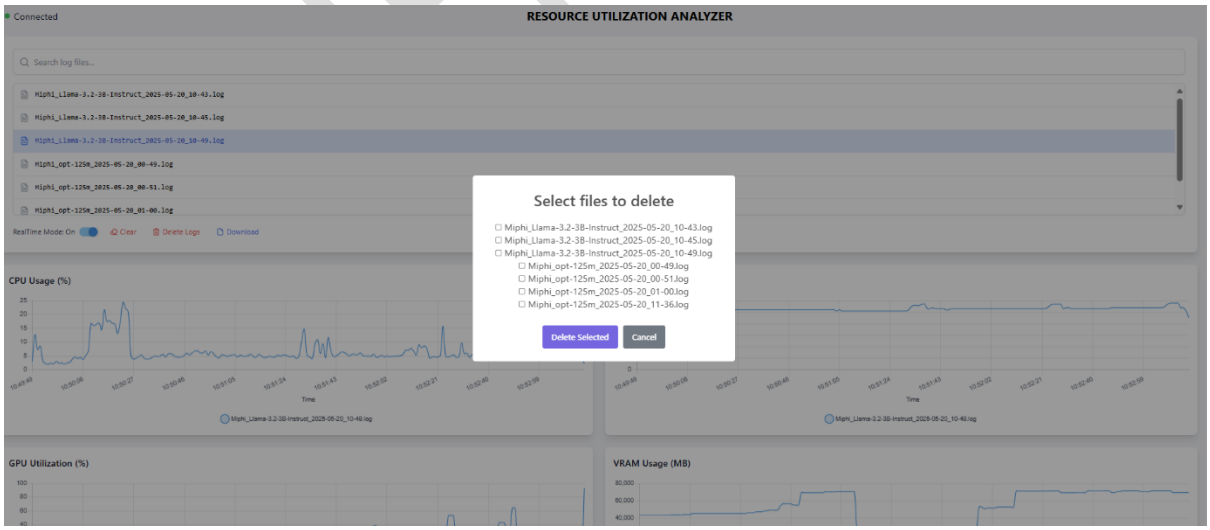


## 12.3 Clear Selection

The **Clear** button allows users to quickly remove all selected log files from the comparison view. This feature helps reset the interface, enabling users to start a new selection without manually deselecting each file.

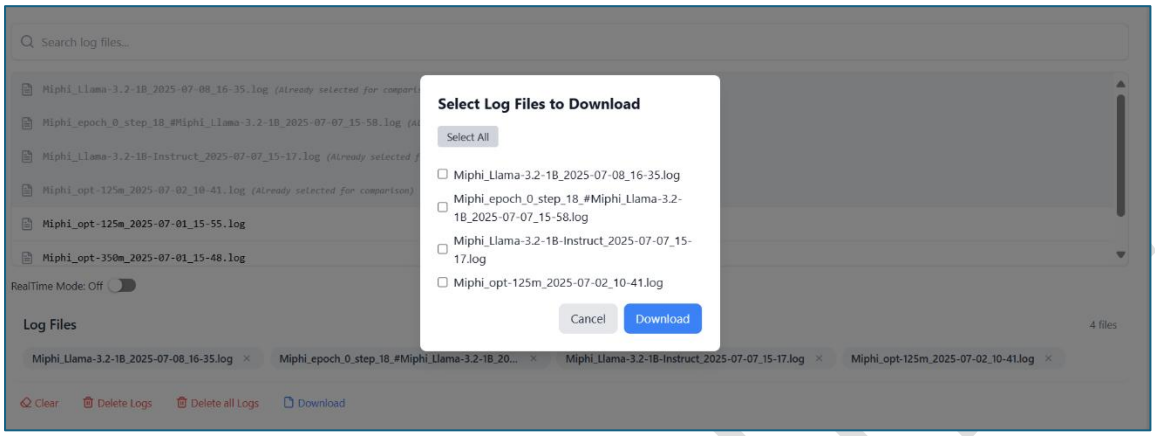
## 12.4 Delete Logs

This section allows users to delete log files individually or in bulk. Users can remove one or more selected log files from the system. This feature enables efficient log management, helping users clean up outdated or unnecessary files without manually deselecting each one.

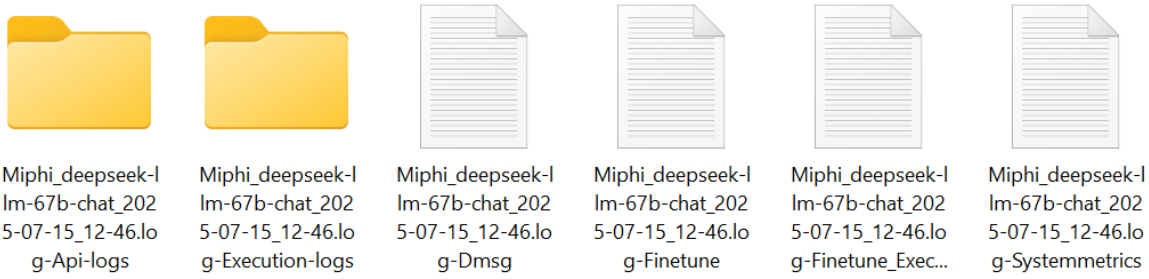


## 12.5 Download Logs

The **Download Logs** button allows users to export detailed log files for further analysis or archival. When a single file is selected, clicking the button downloads a ZIP archive named after the corresponding **Job ID**.




If multiple log files are selected, the interface prompts the user to choose a specific file for download. This ensures precise access to the desired logs without downloading unnecessary data.




**Note**: Downloaded zip folder will contain these files.